European Network of
Transmission System Operators
For Electricity

Installation Guide

Energy Communication Platform
Version 4.16.0

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Energy Communication Platform, version 4.16.0

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2. Executive Summary

This document provides installation instructions for all ECP components.

This document is primarily targeted at Administrators.

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3. Revision History

Version	Date	Author	Description
4.1.4	05.02.2018	Laura Betcherová	Documentation review
4.2.0	23.03.2018	Petr Novák	Automatic start of service after server startup
4.3.0	26.06.2018	Josef Brož	Version update
4.3.0	24.09.2018	Barbora Klabačková	Reverse proxy usage for ECP CD, Endpoint/CD uninstallation via administration console, Java upgrade description
4.3.0	05.10.2018	Barbora Klabačková	Documentation review
4.3.0	18.10.2018	Barbora Klabačková	Documentation review
4.4.0	05.06.2019	Drahoslav Stejskal	External MySQL Database – Create Schema update
4.4.0	05.06.2019	Barbora Kánská	Support forward proxy update
4.5.0	22.08.2019	Jiří Kakrda	Version update
4.6.0	05.11.2019	Jiří Kakrda	Version update
4.6.0	22.11.2019	Barbora Kánská	Document revision, formal changes
4.6.1	18.03.2020	Anna Langrová	Version update
4.7.0	28.06.2020	Jiří Kakrda, Lukáš Voříšek	Changes due to systemctl support
4.7.1	31.07.2020	Anna Langrová	<log_path> description
4.7.2	16.11.2020	Jiří Kakrda	Java installation description
4.8.0	11.03.2021	Tomáš Vondra, Jiří Kakrda	MySQL and PostgreSQL installation, Custom passwords, OpenJPA, Java 11
4.8.1	09.07.2021	Lukáš Voříšek	MSSQL updates

4.8.2	03.12.2021	Daniel Janota, Jiří Kakrda	Windows Server 2019, supported browser update, ActiveMQ configuration files, Registration Process update, Oracle national charset, formatting
4.9.0	26.05.2022	Jiří Kakrda	Supported browser update, ActiveMQ version update, Java version update, GUI login, Broker HA configuration update
4.10.0	07.11.2022	Daniel Janota, Jiří Kakrda	Hibernate, Environmental variables update, Broker ioExceptionHandler, Dashboard screenshot update, CD registration password, Broker configuration update, CentOS 8 deprecation, Oracle 12c and PostgreSQL 9.6 deprecation, Java 11.0.16 disclaimer
4.11.0	16.05.2023	Jiří Kakrda	Extension of Keystore properties description, removal of Linux JAR installation, Java 17, update of supported OSs and DBs, Kubernetes, TLSv1.2 support
4.12.0	27.10.2023	Jiří Kakrda	Removal of info about CD restart after renewal
4.14.0	10.06.2024	Jiří Kakrda	Artemis, CD and Broker Windows removal
4.15.0	21.10.2024	Jiří Kakrda	Windows Server 2022 addition, Windows Server 2016 removal
4.16.0	06.05.2025	Jiří Kakrda	Self-signed certificates error page info, HA installation chapters order, SQL Server 2022, MySQL 8.4, PostgreSQL 16, Artemis external database

4. Service Level Requirements

Please note that the following HW requirements are only estimations, and they may be changed according to performance requirements. In general, we recommend the usage of virtual machines, which can be resized when more resource is necessary.

Real HW requirements also depend on message traffic.

4.1. Recommended Minimum Hardware Requirements

4.1.1. Endpoint with Internal Broker

For an Endpoint, which will at peak send a few thousands of messages per hour, we recommend the following configuration.

CPU	4 cores – each at least at 2Ghz
RAM	8 GB
HDD	For ECP: 40 GB, For Message Archive: depends on the message traffic

4.1.2. Broker

For a Broker, which will exchange messages between 50 Endpoints, each sending at peak few thousands of messages per hour, we recommend the following configuration.

CPU	8 cores – each at least at 2Ghz
RAM	16 GB
HDD	100 GB

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4.1.3. Component Directory

To support network of up to 50 Endpoints, we recommend following hardware configuration for Component Directory.

CPU	2 cores – each at least at 2Ghz
RAM	4 GB
HDD	10 GB

4.1.4. Docker

To support network of CD, Broker and up to 10 Endpoints (each with own Internal Broker), we recommend following minimum hardware configuration for the ECP network running in Docker.

CPU	8 cores – each at least at 2Ghz
RAM	32 GB
HDD	100 GB

4.2. Supported Operating Systems

- *Windows Server 2019*
- *Windows Server 2022*
- *RHEL 8*
- *RHEL 9*
- *Oracle Linux 8*
- *Oracle Linux 9*

4.3. Supported Web Browsers

Google Chrome 133.0.6943.142 (64-bit), Firefox 135.0.1, Safari 18.3, MS Edge 133.0.3065.92

4.4. Supported Resolution

Minimal supported resolution is 1280x1024.

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5. General Installation Prerequisites

5.1. Java

- *Java 17 is required. Recommended Java provider is Red Hat.*
- *Minimal Java version compatible is 17.0.14.*
- *Check the current Java version using the command:*
  - ☐ *java -version*
- *Expected output:*
  - *openjdk version "17.0.14" or newer versions*
- *If the version is below 17.0.14 or if the 0.14-update is more than 6 months old:*
  - ☐ *Download and install the latest Java 17 for your OS.*
  - *WINDOWS*
    - ☐ *JRE 17 installation package: https://developers.redhat.com/products/openjdk/download*
    - ☐ *After installation, configure the JRE_HOME system variable.*
    - ☐ *Open File Explorer*
    - ☐ *Right click on This PC and select Properties*
    - ☐ *Select Advanced system settings -> Environment Variables -> System Variables -> New…*
      - *Name: JRE_HOME*
      - *Value: path to Java 17 (e.g. C:\Program Files\RedHat\ java-17-openjdk-17.0.14.0.7-1.jre\)*
    - ☐ *Make sure that the JAVA_HOME system variable is not configured in Advanced system settings -> Environment Variables -> System Variables*
      - *If JAVA_HOME system variable is set, remove it using the Delete button*
  - *LINUX*
    - ☐ *Using Unix repositories – yum install java-17-openjdk*
- *After Java installation, configure Java properties:*

> ☐ *Open the security configuration file, e.g. on path: /usr/lib/jvm/ java-17-openjdk-17.0.14.0.7-2.0.1.el9.x86_64/conf/security/java.security*

> ☐ *Find the line securerandom.source=file:/dev/random*

> ☐ *Change the line to securerandom.source=file:/dev/urandom*

> ☐ *Save the file*

5.2. Firewall

When the Firewall between servers where ECP Components are installed is present, before processing with the installation, there is a need to open following flows on Firewall to enable communication between components. Please note that the following ports are set by default, and you need to check your current configuration of used ports.

5.2.1. Default Ports

- *Endpoint: 8443*
- *Internal Broker: 5672, 8161*
- *Component Directory: 8443*
- *Broker: 5671, 8161*

5.2.2. Command Line Configuration on Linux

Firewall must be disabled or configured to enable ports, e.g. 8443:

- *firewall-cmd --permanent --add-port=8443/tcp*
- *firewall-cmd --reload*

The following command should show port 8443 among allowed ports:

- *firewall-cmd --list-all*

5.2.2.1. Extension of Firewall for Loadbalancer (cluster)

When deploying cluster, port 8009 should be enabled:

- *firewall-cmd --permanent --add-port=8009/tcp*
- *firewall-cmd --reload*

5.2.3. GUI Configuration on Windows

Firewall must be disabled or configured to enable ports, e.g. 8443:

- *Go to Control Panel – System and Security – Windows Firewall*

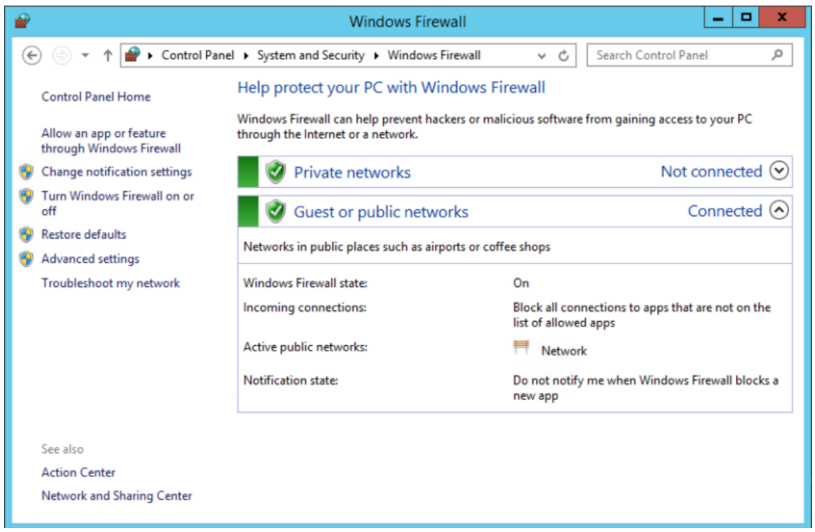

Figure 1: Windows Firewall

- *Click on Advanced Settings and go to Inbound Rules*

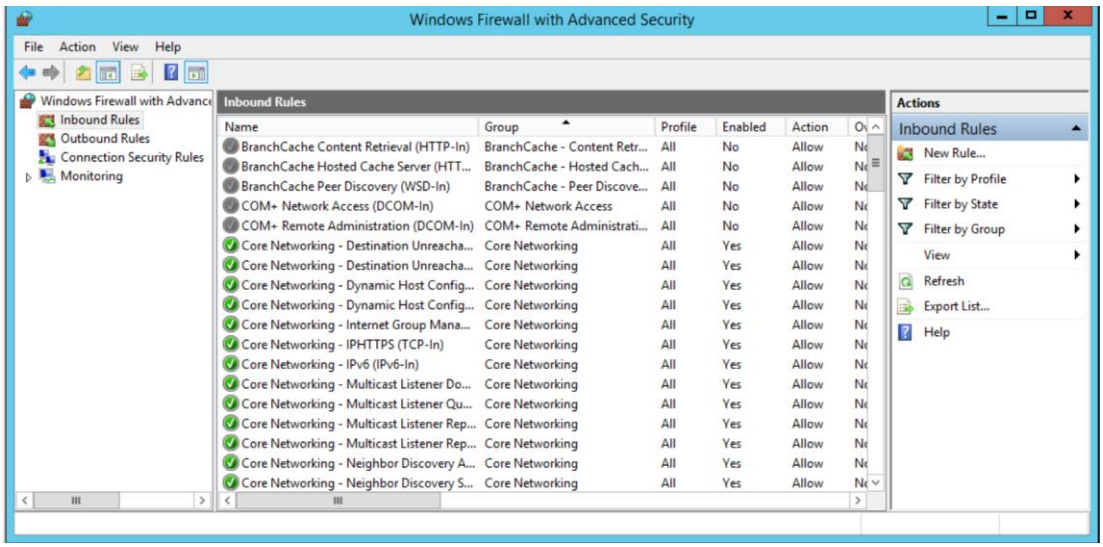

Figure 2: Windows Firewall - Inbound Rules

- *Click on New Rule and specify it as: Port – TCP + Specific local ports: 8443 – Allow the connection – Domain/Private/Public according to your default settings – desired Name – Finish*
- *New rule is displayed in the list of Inbound Rules*

5.3. Databases

When Firewall between servers where Databases are situated is present, before processing with the installation, there is a need to allow following connections between ECP and Databases. Please note that the following ports are set by default, and you need to check your current configuration of used ports. Note: Does not apply to embedded DB.

5.3.1. Default Ports

- *Oracle: 1521*
- *MSSQL: 1433*
- *MySQL: 3306*
- *PostgreSQL: 5432*

5.4. Component Code (EIC V-Code)

Before processing with the installation, please prepare the specific component code for registration of your component. Component code must be unique (within each ECP network). The ECP Network Administrator should choose an appropriate coding schema that ensures uniqueness even in case of connected multiple ECP Networks.

In the Energy sector, the Energy Identification Coding (EIC) scheme is mostly used. See https://www.entsoe.eu/data/energy-identification-codes-eic/ for more details.

There is EIC Object type V (so-called EIC V-Code) used for identification of the locality. The ECP components may use EIC V-Codes in the Endpoint function as the Component Codes.

Valid EIC V-code is issued by LIO/CIO (Local Issuing Office/ Central Issuing Office). Contact LIO for your country to request a new EIC V-Code. The list of LIOs is available on the ENTSO-E Website under the link mentioned above.

5.5. ECP Installers Checksum Verification

ECP is provided with ***ecp-checksum-<ecp-version>.txt*** file, which includes checksums of all ECP installers files. Use this file to verify ECP installers data integrity.

To verify digital signature of the checksum file, you will also need to download ***ecp-checksum-<ecp-version>.sig*** signature file and ***ecpedx_signing_public.pem*** file with a public key. Then you can verify the signature with the following command:

- *openssl dgst -sha512 -verify ecpedx_signing_public.pem -signature ecp-checksum-<ecp-version>.sig ecp-checksum-<ecp-version>.txt*

A *"Verified OK"* response message should appear if the file verification is successful.

6. General Information

6.1. ECP Introduction

6.1.1. ECP Network Overview

The ECP network consists of multiple ECP Endpoints, Component Directories, and Brokers. Each Endpoint must be connected to Internal broker. From the point of ECP network, Endpoint and Internal Broker are considered as one component. Internal Broker does not use the component code, and it is not registered to Component Directory.

This guide is focused on installation of these components.

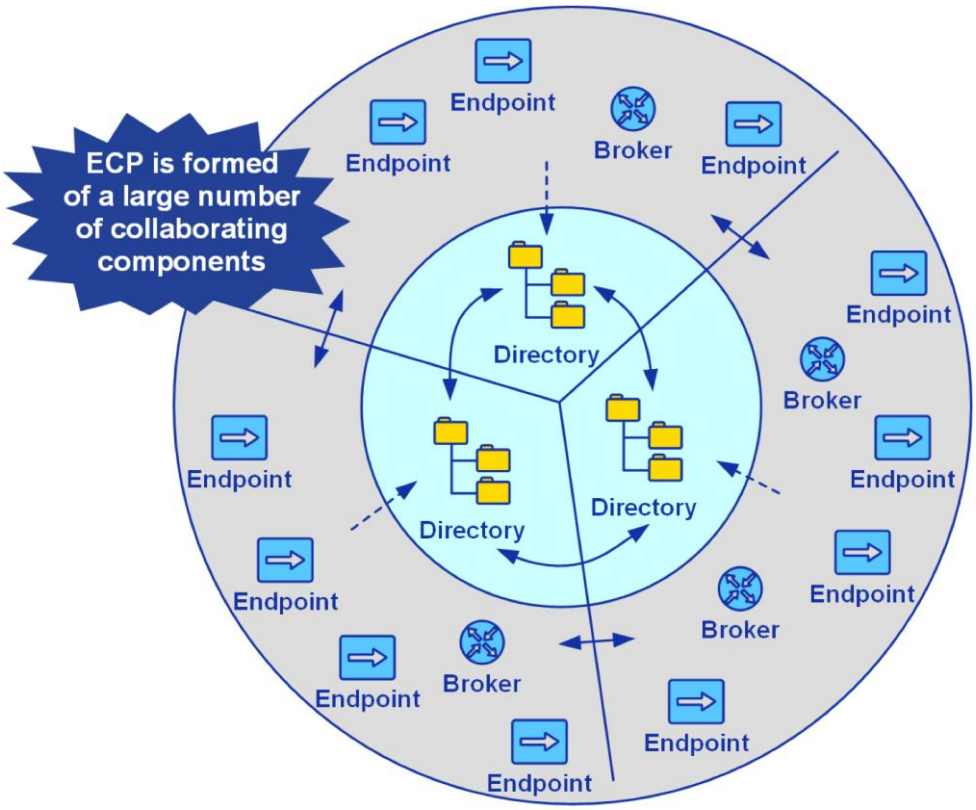

Figure 3: Distributed architecture

6.1.2. ECP Network Communication Protocols

ECP uses two main protocols for messaging – AMQPS and HTTPS. Please note that the port numbers specified below are default, however different ports may be explicitly configured for ECP depending on the installation environment.

Following is the list of all ports that must be open to allow ECP communication:

6.1.2.1. ECP Endpoint

WAN

> *Outgoing*
  - *5671 - to ECP Brokers*
  - *5671 - to Internal Brokers with enabled direct communication (only if direct communication is enabled)*
  - *8443 - to ECP Component Directories*

Internal network

> *Incoming*
  - *5672 – AMQP API (only when the AMQP channel is enabled)*
  - *8443 – Webservice API, GUI and monitoring*

> *Outgoing*
  - *<db-port> – optionally to the selected external database in HA deployment*

6.1.2.2. Internal Broker

WAN

> *Incoming*
  - *5671 – only when direct communication is enabled*

Internal network

> *Incoming*
  - *5672 – AMQPS port for messaging*
  - *8161 – Management GUI and monitoring*

6.1.2.3. ECP Component Directory

WAN

> *Outgoing*
  - *8443 – to other ECP Component Directories in the network*

> *Incoming*

- *8443 – synchronization REST API exposed for other components*

Internal network

> *Incoming*

- *8443 – GUI and monitoring*

> *Outgoing*

- *<db-port> – optionally to the selected external database in HA deployment*

6.1.2.4. ECP Broker

WAN

> *Outgoing*

- *8443 - to ECP Component Directories*

> *Incoming*

- *5671 – AMQPS port for messaging*

Internal network

> *Incoming*

- *8161 – Management GUI and monitoring*

6.1.3. ECP Network Deployment Modes

The ECP components can be installed using several modes. Before starting an installation, it is important to choose one of the following installation modes:

- *Standalone deployment*
- *High availability (HA) deployment*
- *Clustered high availability deployment*

6.1.3.1. Standalone Deployment

In standalone mode, no other services are required (such as a database, however it is still possible to configure use of an external database). This is the simplest installation option. Please note that high availability is not possible in this mode.

6.1.3.2. High Availability Deployment

High availability deployment consists of a shared external database and two ECP component deployments. In this mode, when one ECP component goes down, the system will continue to function. Please note that for HA deployment two IPs in WAN are required in the following scenarios:

- *ECP Broker*
- *Internal Broker with enabled direct communication*

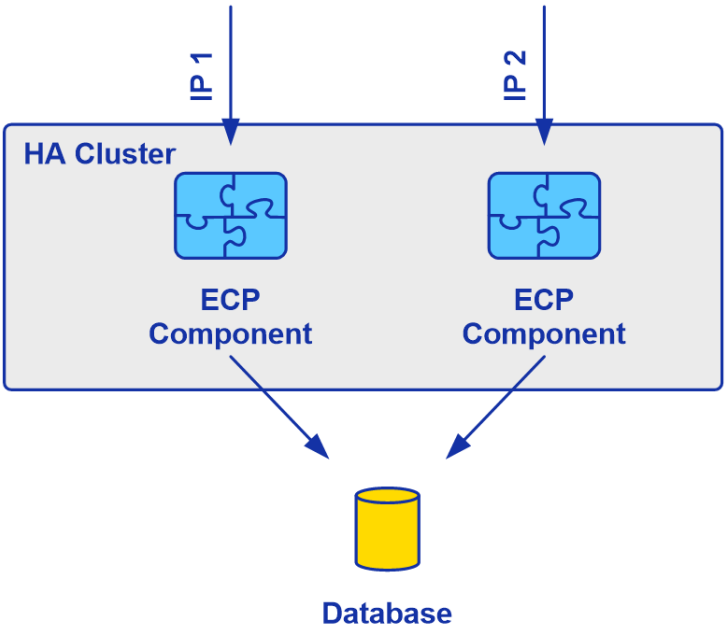

Figure 4: HA deployment

If you also want to have HA at the database level it is possible to deploy MySQL to the cluster using Galera cluster or by using other solution that works for your database.

6.1.3.3. Clustered High Availability Deployment

Clustered high availability deployment is possible for Endpoints that are required to send and receive so many messages that two application servers do not have enough performance to process them.

Clustered HA deployment consists of:

- *N application servers with ECP Endpoint and Internal Broker installed*
    - *ECP Endpoint and Internal Broker can be in the same server, or in separate servers*
- *An external database (e.g. MySQL)*
- *Shared file storage*

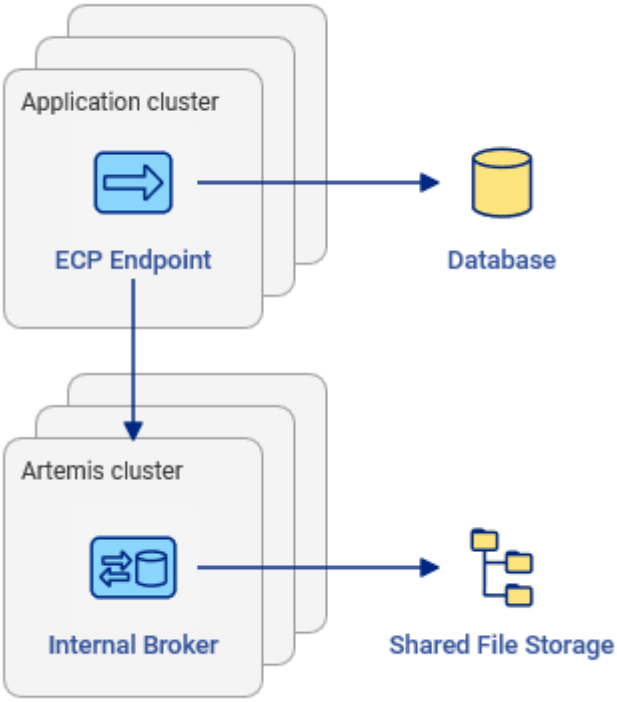

Figure 5: Clustered HA deployment

7. Internal Broker Installation

When High Availability mode is required, first proceed with either *Installation on Windows* or *Installation on Linux* chapter (based on the host operating system), then proceed with *Installation in High Availability mode* chapter.

7.1. Common Prerequisites

7.1.1. General Prerequisites Before Installing Internal Broker

- *Internal Broker installer (jar or rpm package, depending on the deployment and OS type)*
- *Before processing, please check the General Installation Prerequisites.*

7.2. Installation on Windows

The application may be installed in either command line or GUI mode. Both modes are described in this chapter.

7.2.1. Variables used in the guide

Several variables are used throughout this installation guide using the <variable_name> format. These variables are used only within this guide and are not to be confused with variables of your OS.

Name	Description
<jre_install_path>	Location of the JRE 17 used by the application
<install_path>	Path pointing to the target installation directory
<version>	Arbitrary version name

<eccosp-artemis-jar> Path to the Internal Broker installation package

7.2.2. Prerequisites

Download the Internal Broker .jar file (*eccosp-artemis*) and copy it to the server.

Before processing, please check the *General Installation Prerequisites*.

Please make sure that MMC (Microsoft Management Console) or Services snap-in is not running during the installation as it prevents successful registration of the ECCo SP service.

7.2.3. Installation

7.2.3.1. GUI Installation

1. ☐ *Double-click the <eccosp-artemis-jar> file. If Java was registered to execute jar files, an installer window should open. In case of an issue, please use the command line installation described in chapter Command line installation.*
2. *Internal Broker installer window should open.*

- ☐ *Select installation type 'Install' to install the Internal Broker application as a standalone application.*
- ☐ *Select target installation path (the target directory will be created)*

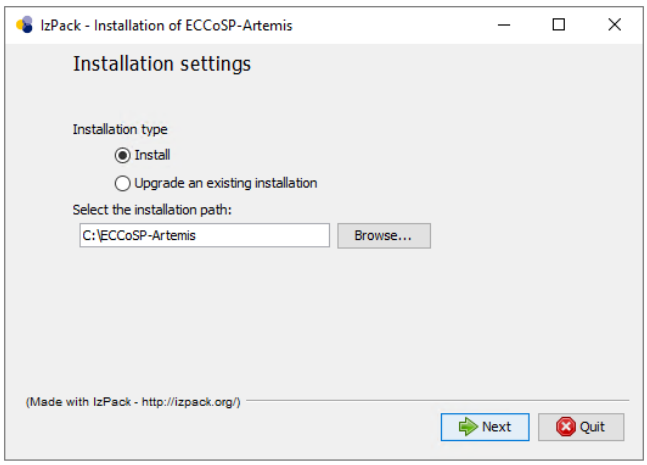

Figure 6: Installation 1

- ☐ *Click Next*
- ☐ *Choose whether the Internal Broker application should be installed as a Windows service.*

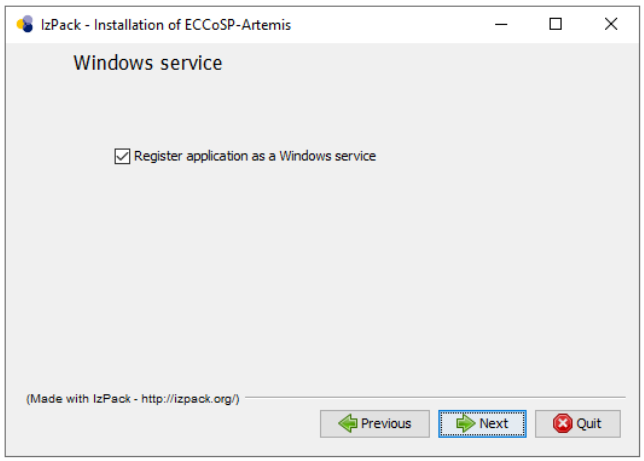

Figure 7: Installation 2

- ☐ *Click Next*
- ☐ *Choose whether the Internal Broker application should be automatically started after server start. The preferred option is to not start Internal Broker after server start.*

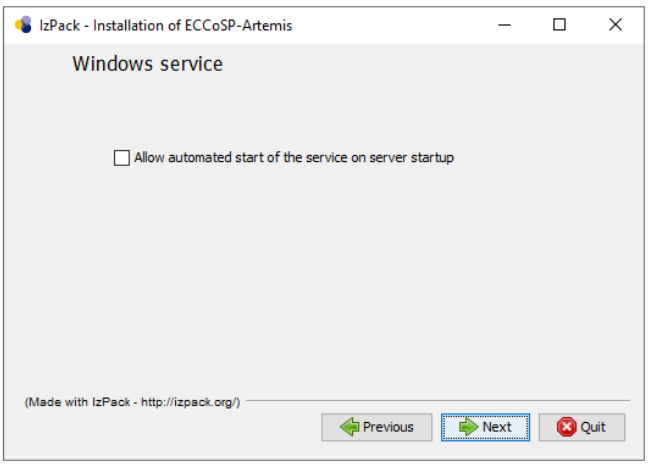

Figure 8: Installation 3

- ☐ *Click Next*
- ☐ *Application is installed*
- ☐ *Choose whether to use a default password ("password") in all configuration files and keystores or to define custom passwords.*

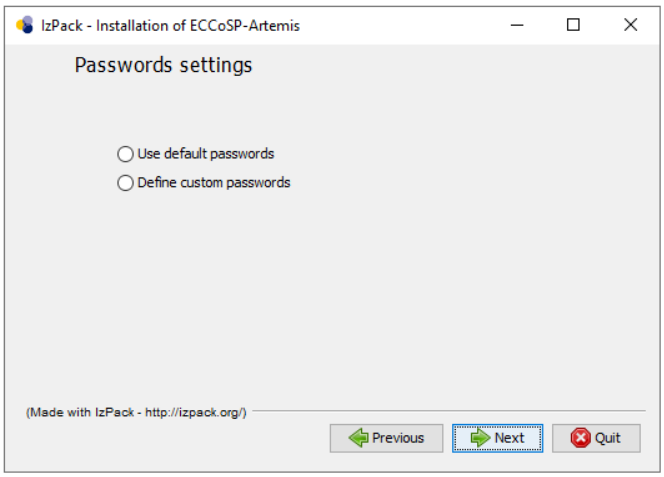

Figure 9: Installation – Password Selection

- ☐ *If custom passwords are selected, detailed option for custom settings will appear.*
    - *Plain – User password is stored in a plain form in a properties file*
    - *Hashed – User password is hashed*
    - *Encrypted – Keystore passwords is encrypted*

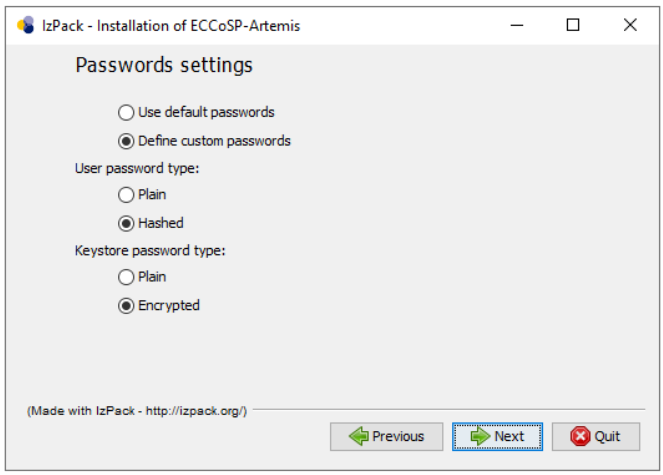

Figure 10: Installation – Password Settings

- ☐ *Click Next*
- ☐ *Passwords screen – type in user and keystore (JKS keystore used for AMQPS connections) passwords*

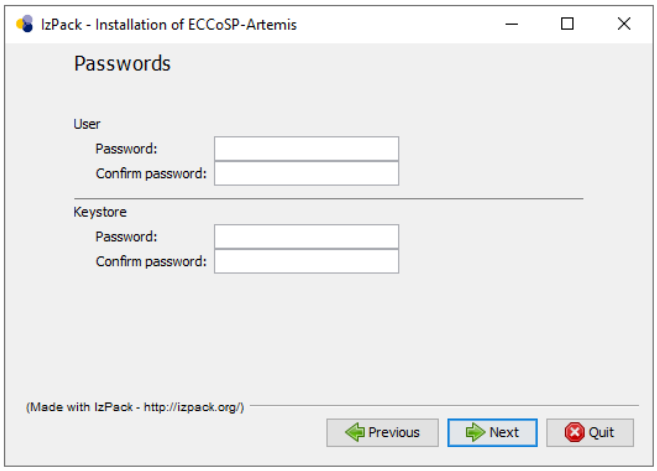

Figure 11: Installation – Passwords

- ☐ *Click Next*
- ☐ *Custom passwords are configured*
- ☐ *If desired, an automatic installation script can be generated. It can be used in the future to install the application using the same options.*

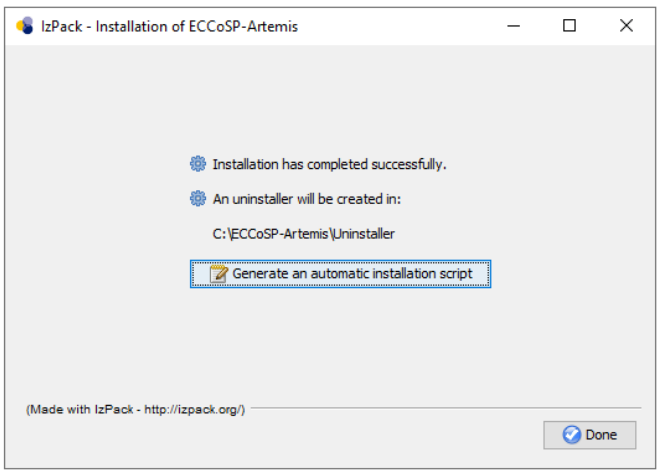

Figure 12: Installation 4

- ☐ *Click Done*

3. ☐ *Configure Internal Broker (see chapter Internal Broker configuration)*

7.2.3.2. Command line installation

Open command line with the administrator rights:

- ☐ *Start > type "cmd" > right click > run as administrator*
- ☐ *Change directory to location of <eccosp-artemis-jar>, e.g.*
  - *cd c:\eccosp-artemis*

- ☐ *Run the <eccosp-artemis-jar> file using*
  - *java -jar <eccosp-artemis-jar> -console*
- ☐ *Follow the console installation wizard*

Then configure Internal Broker before starting it (see chapter *Internal Broker configuration*).

7.2.4. Starting and stopping Internal Broker

7.2.4.1. Using GUI

a) If installed as a service:

- *Start > Run > "services.msc"*
- *Find service with name "eccosp-artemis"*
- *Start / stop service by buttons on the top left*

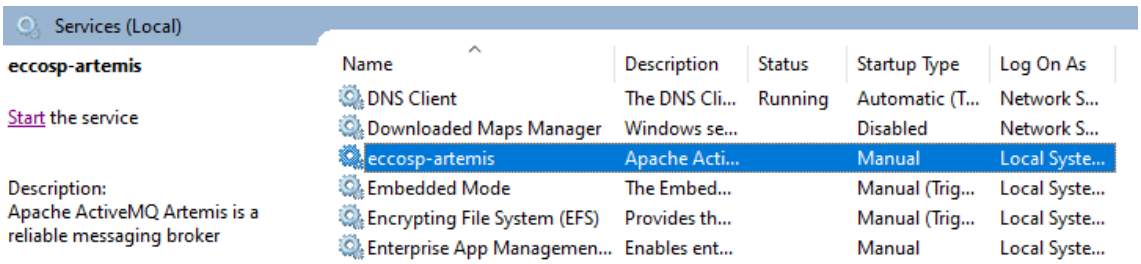

Figure 13: Services

b) If not installed as a service:

- *Start application by using*
  - <install_path>\broker\bin\artemis.cmd run
- *Stop application by using*
  - <install_path>\broker\bin\artemis.cmd stop

7.2.4.2. Using Command line

a) If installed as a service:

- *Start service using*
  - *SC start eccosp-artemis*
- *Stop service using*
  - *SC stop eccosp-artemis*

b) If not installed as a service:

- *Start application by using*

  - <install_path>\broker\bin\artemis.cmd run

- *Stop application by using*
  - <install_path>\broker\bin\artemis.cmd stop

7.2.5. Installation verification

The application should be installed in the <install_path> folder. This installation folder should contain Artemis core files, broker instance folder, and an uninstaller.

7.2.5.1. Service status

The status of the service, if installed, can be checked either via command line:

- *SC query eccosp-artemis*

or in the Windows Services tool.

7.2.5.2. Log files

Check the application log files for more information about its status. The default location is *<install_path>\broker\log*.

7.2.5.3. GUI

GUI of Artemis Broker is started on port 8161. By default, the access is limited for opening it on the server where the Artemis is installed.

For allowing connection to GUI from other locations it is necessary to change the configuration file *<install_path>\broker\etc\jolokia-access.xml.*

> *Add <allow-origin> element with an URL pattern to match URLs from where the access will be allowed*

> *Multiple <allow-origin> elements can be defined*

> *URL pattern supports a wildcard character: **

Also, it is necessary to specify the URL in the *<install_path>\broker\etc**bootstrap.xml***:

> *In the binding element, find and change the uri attribute:*

  - *E.g. uri="http://0.0.0.0:8161"*
  - *or*
  - *uri="http://<server IP address>:8161"*

Please note that the Artemis Prometheus metrics plugin, which exposes the metrics at **/metrics* URL, does not support the authentication. If you wish to enable a remote access to the console without exposing the metrics, it is possible to disable the metrics in *<install_path>\broker\etc**bootstrap.xml***:

> *In the binding element, find and remove this line:*
  - *<app url="metrics" war="metrics.war"/>*

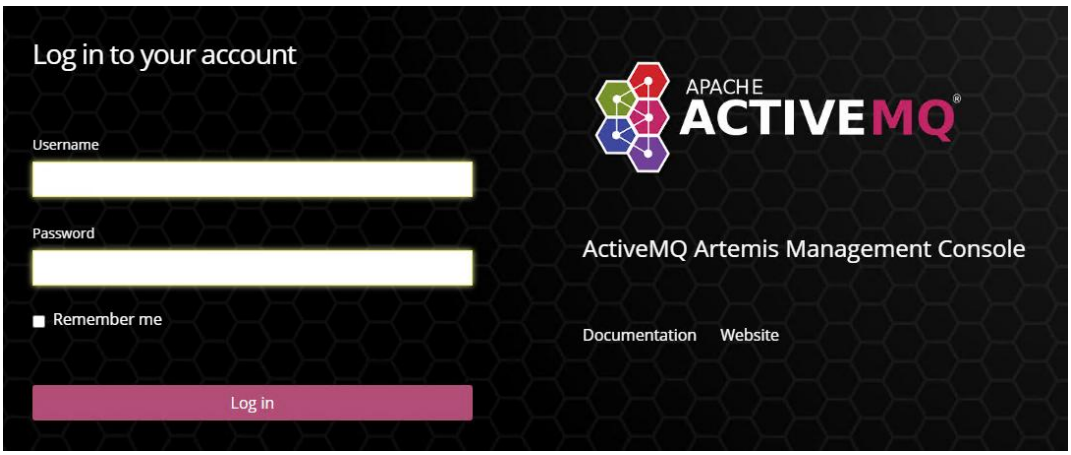

Figure 14: GUI – Verification

7.3. Uninstallation on Windows

Stop the application and run the command line as an administrator. Navigate to *<installation directory>\Uninstaller* and execute the following command:

- *java -jar uninstaller.jar -c*

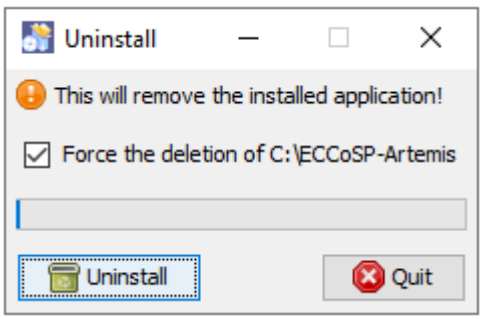

Figure 15: Uninstallation 1

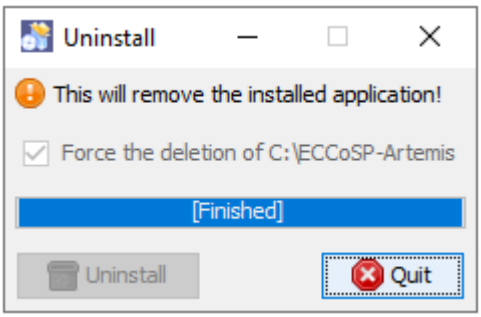

Figure 16: Uninstallation 2

Uninstaller will remove all application components and unregister the service if present.

7.4. Installation on Linux

The following section describes the installation of Internal Broker on the Linux platform.

7.4.1. Variables used in the guide

Several variables are used throughout this installation guide using the <variable_name> format. These variables are used only within this guide and are not to be confused with variables of your OS.

Name	Description
<jre-rpm>	Path to the JRE 17 installation package
<jre_install_path>	Location of the JRE 17 used by application
<eccosp-rpm>	Path to the Internal Broker installation package
<version>	Arbitrary version name

7.4.2. Prerequisites

Download the Internal Broker rpm file (*eccosp-artemis*) and copy it to the server.

Before processing, please check the *General Installation Prerequisites*.

Please note that ECCo SP rpm file has a dependency to specified by jre therefore installed Java must provide jre. Java headless OpenJDK distribution can be also used, however it will not pass the rpm dependency check. To overcome this issue, install rpm with nodeps option.

7.4.3. Installation

Change the directory to the folder where the ECCo SP rpm installer is located and then invoke rpm install.

- ☐ *rpm -i <eccosp-rpm>*

By default, Internal Broker is set up to automatically start after server start. The automatic start can be disabled by the following command:

- ☐ *systemctl disable eccosp-artemis*

Installer automatically creates the following folder structure:

Folder	Description
/usr/share/eccosp-artemis	Folders for Artemis core files
/opt/eccosp-artemis	Folders for Artemis broker instance
/opt/eccosp-artemis/etc	Artemis broker instance configuration files
/opt/eccosp-artemis /data	Artemis broker instance data

7.4.4. Starting and stopping Internal Broker

After Internal Broker is configured, it can be started. Internal Broker is registered as a Linux service. The following commands are available:

- *To start Internal Broker, use command "systemctl start eccosp-artemis.service"*
- *To stop Internal Broker, use command "systemctl stop eccosp-artemis.service"*
- *To restart Internal Broker, use command "systemctl restart eccosp-artemis.service"*

If not installed as a service:

- *Start application by using "/opt/eccosp-artemis/bin/artemis run"*
- *Stop application by using "/opt/eccosp-artemis/bin/artemis stop"*

7.4.5. Installation verification

Check the status of the eccosp-artemis service:

- *systemctl status eccosp-artemis.service*
- *The service should be running*

Display application log (**artemis.log**) output:

- *tail /opt/eccosp-artemis/log/artemis.log*

GUI of Artemis Broker is started on port 8161. By default, the access is limited for opening it on the server where the Artemis is installed.

For allowing connection to GUI from other locations it is necessary to change the configuration file */opt/eccosp-artemis/etc/**jolokia-access.xml**.*

> *Add <allow-origin> element with an URL pattern to match URLs from where the access will be allowed*

> *Multiple <allow-origin> elements can be defined*

> *URL pattern supports a wildcard character: **

Also, it is necessary to specify the URL in the */opt/eccosp-artemis/etc/**bootstrap.xml***:

> *In the binding element, find and change the uri attribute:*

- *E.g. uri="http://0.0.0.0:8161"*

  or

- *uri="http://<server IP address>:8161"*

Please note that the Artemis Prometheus metrics plugin, which exposes the metrics at **/metrics* URL, does not support the authentication. If you wish to enable a remote access to the console without exposing the metrics, it is possible to disable the metrics in */opt/eccosp-artemis/etc/**bootstrap.xml***:

> *In the binding element, find and remove this line:*

- *<app url="metrics" war="metrics.war"/>*

7.4.6. Log files

Internal Broker log files are available in the */opt/eccosp-artemis/log* folder.

7.4.7. Application accounts

RPM Internal Broker installation automatically creates the user account *eccosp-artemis* with UID 2030 that belongs to the *eccosp-artemis* group with GID 2030. Please note that it is not possible to change the UID and GID, so make sure that there is no other user using the same IDs.

7.4.8. Custom passwords

Default user and keystore are delivered with default passwords. To define custom passwords, which will replace default passwords, run the *configure_passwords.sh* script in */opt/eccosp-artemis/etc*.

After starting the script:

- ☐ *Select the desired form of the user password*
    - *Plain – User password is stored in plain form in artemis-users.properties file*
    - *Hashed – User passwords is hashed*
- ☐ *Select the desired for of the keystore password*
    - *Plain – Keystore password will be stored in plain form in broker.xml file*
    - *Encrypted – Keystore password is encrypted*
- ☐ *Fill the user and keystore custom passwords:*

7.5. Uninstall on Linux

To uninstall Internal Broker, simply run the following command:

- ☐ *rpm -e eccosp-artemis*

7.6. Installation in High Availability mode

Internal Broker can be deployed in HA mode on two or more application servers. This setup consists of the following infrastructure components:

- *Two or more application servers with Internal Broker.*
- *Shared file storage.*

If you wish to use an external database instead of the shared file storage, please refer to the Artemis documentation.

7.6.1. Artemis HA limitations in Windows

In the Windows environment, Artemis has few limitations in supporting various network file sharing protocols, unlike Linux where this isn't an issue. While Artemis supports NFSv4, Windows only fully supports this protocol version in its Server version, leaving the Client version restricted to v3.

For setting up Artemis HA in Windows, three potential approaches are available:

- *Single Artemis Node with ECCo SP Components in HA*
    - *Maintain a single Artemis node, configuring only the connected ECCo SP components for high availability. However, this leaves the Artemis component vulnerable as a single point of failure.*
- *Data Replication Strategy*
    - *Implement a data replication strategy instead of relying on shared storage. Please refer to the Artemis documentation of the data replication.*
- *Use Docker for Windows*
    - *Use Docker for Windows, with each Windows server running a Docker instance housing an Artemis container. The shared storage folder can be directly mounted from the Artemis container using the NFSv4 Client. The folder could be shared by the Windows server itself, as it supports NFSv4 (only in the role of Server, not the Client).*

7.6.2. Setup application servers

Please follow the installation procedure. Only one server is needed for initial setup.

7.6.3. Configure application

Open ***broker.xml*** in the following folder:

- *Windows:*
    - *<install_path>\broker\etc*
- *Linux:*
    - */opt/eccosp-artemis/etc*

Configure a path to the shared storage, which is accessible by all nodes, in these elements:

- *<paging-directory>*
- *<bindings-directory>*
- *<journal-directory>*
- *<large-messages-directory>*

Find and uncomment the *<connectors>* element.

- *Inside the element, keep only one connector with a name according to the node type: primary (master), or backup (slave)*
- *Configure the connector IP and port. It must be accessible from other nodes.*

Uncomment the *"artemis"* acceptor in the *<acceptors>* element.

- *Configure the acceptor port to match the port in the connector.*

Find and uncomment the *<cluster-user>* and *<cluster-password>* elements.

- *Change the default values. User and password must be the same in all nodes.*

Find and uncomment the *<cluster-connections>* element.

- *Inside the element, keep only one connector-ref and static-connectors according to the node type:*
    - *Primary (master)*

```
<connector-ref>master</connector-ref>
<static-connectors>
    <connector-ref>slave</connector-ref>
</static-connectors>
```

    - *Backup (slave)*

```
<connector-ref>slave</connector-ref>
<static-connectors>
    <connector-ref>master</connector-ref>
</static-connectors>
```

Find and uncomment the *<ha-policy>* element.

- *Inside the element, keep only <master> or <slave> element according to the node type.*

For more details about the Artemis configuration, please refer to the official Artemis documentation. ECCo SP Internal Broker does not alter the Artemis High Availability behaviour, so any Artemis configuration will work with Internal Broker.

7.6.3.1. NFS Mount Recommendations

When using NFS for the shared storage, NFSv4 version must be used.

Please refer to the official Artemis documentation for recommendations on the configuration options to improve the reliability.

7.7. Internal Broker configuration

7.7.1. Keystore configuration

By default, Internal Broker accepts connections on 5672 port, and verifies AMQPS connections by using a provided default keystore, which is included in *keystore.jks* in the following path:

- *Windows:*
  - *<install_path>\broker\etc*
- *Linux:*
  - */opt/eccosp-artemis/etc*

It is recommended to replace the default keystore with a custom one. Either replace the *keystore.jks*, or change properties in the *acceptor* in ***broker.xml*** in the configuration folder:

- *Windows:*
  - *<install_path>\broker\etc*
- *Linux:*
  - */opt/eccosp-artemis/etc*

Change the following properties:

- *keyStorePath*
- *keyStorePassword*
- *trustStorePath*
- *trustStorePassword*

When a keystore contains multiple keypairs, it is necessary to add the *keyStoreAlias* property referencing a keypair which should be used.

When Internal Broker does not use a default keypair, its public certificate must be imported to the authKeystore.jks of all connected Endpoints. Connected Business Applications must also use the Internal Broker public certificate.

7.7.2. Journal buffer timeout

Artemis maintains an internal buffer for the file journal instead of writing directly to the journal with every change. This buffer is flushed when it is full, or when a timeout expires.

Artemis provides a script for detecting an optimal timeout value:

- *Windows:*
  - *<install_path>\broker\bin\artemis perf-journal*
- *Linux:*
  - */opt/eccosp-artemis/bin/artemis perf-journal*

Take the returned value and replace a default value in the *journal-buffer-timeout* element in ***broker.xml*** in the configuration folder:

- *Windows:*
  - *<install_path>\broker\etc*
- *Linux:*
  - */opt/eccosp-artemis/etc*

7.7.3. ECP Endpoint component code in audit log

To log a correct code of the connected ECP Endpoint in the Internal Broker audit log, open ***broker.xml*** file in the following path:

- *Windows:*
  - *<install_path>\broker\etc*
- *Linux:*
  - */opt/eccosp-artemis/etc*

Find the *broker-plugins* element and replace *CHANGE-ME-COMPONENT-CODE* with a component code of the connected ECP Endpoint, or with a list of component codes in case that several ECP Endpoints are connected to the same Internal Broker, or with some other specific name (e.g. central-internal-broker). Please note, that the Internal Broker's component code serves only as an identification within the audit log and chosen code will not affect the broker functionality.

7.7.4. TLSv1.2 support

By default, Internal Broker supports only TLSv1.3 protocol for incoming connections. If there is a Business Application which cannot use TLSv1.3, then TLSv1.2 must be manually enabled in the configuration files.

If the Business Application connects to the Internal Broker AMQPS protocol, open ***broker.xml*** file in the following path:

- *Windows:*
  - *<install_path>\broker\etc*
- *Linux:*
  - */opt/eccosp-artemis/etc*

- ☐ *Find the acceptor element with "amqps-internal" name*
- ☐ *Set "TLSv1.2,TLSv1.3" in the enabledProtocols attribute*
- ☐ *Set these ciphers in the enabledCipherSuites attribute:*
  - *TLS_AES_256_GCM_SHA384,TLS_CHACHA20_POLY1305_SHA256,TLS_AES_128_GCM_SHA256,TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,TLS_DHE_RSA_WITH_AES_128_GCM_SHA256*

8. ECP Endpoint Installation

When High Availability mode is required, first proceed with either *Installation on Windows* or *Installation on Linux* chapter (based on the host operating system), then proceed with *Installation in High Availability mode* chapter.

8.1. Common Prerequisites

8.1.1. General Prerequisites Before Installing ECP Endpoint

- *ECP Endpoint installer (jar or rpm package, depending on the deployment and OS type)*
- *Registration keystore (this keystore will be provided by your home Component Directory administrator)*
- *Before processing, please check the General Installation Prerequisites.*

8.2. Installation on Windows

The application may be installed in either command line or GUI mode. Both modes are described in this chapter.

8.2.1. Variables used in the guide

Several variables are used throughout this installation guide using the <variable_name> format. These variables are used only within this guide and are not to be confused with variables of your OS.

Name	Description
<jre_install_path>	Location of the JRE 17 used by the application
<install_path>	Path pointing to the target installation directory
<version>	Arbitrary version name
<ecp-jar>	Path to the ECP Endpoint installation package

8.2.2. Prerequisites

Download the ECP Endpoint .jar file and copy it to the server.

Before processing, please check the *General Installation Prerequisites*.

Please make sure that MMC (Microsoft Management Console) or Services snap-in is not running during the installation as it prevents successful registration of the ECP service.

8.2.3. Installation

8.2.3.1. GUI Installation

4. ☐ *Double-click the <ecp-jar> file. If Java was registered to execute jar files, an installer window should open. In case of an issue, please use the command line installation described in chapter Command line installation.*
5. *ECP Endpoint installer window should open.*

- ☐ *Select installation type, whether ECP Endpoint application should be installed as a standalone application or should be installed to an existing application server.*
- ☐ *Select target installation path (the target directory will be created)*

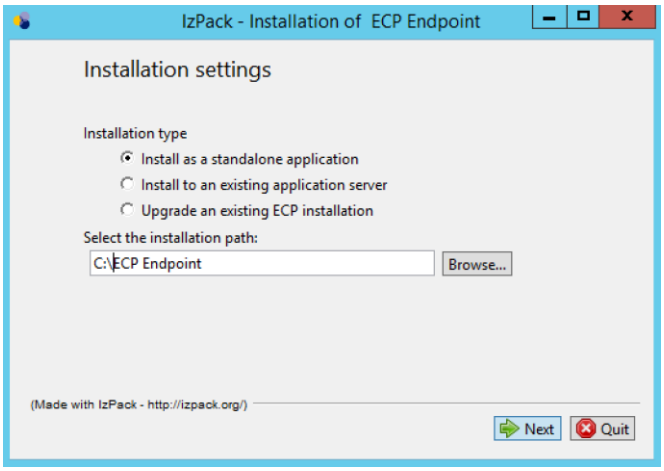

Figure 17: Installation 1

- ☐ *Click Next*
- ☐ *Choose whether the ECP Endpoint application should be installed as a Windows service.*

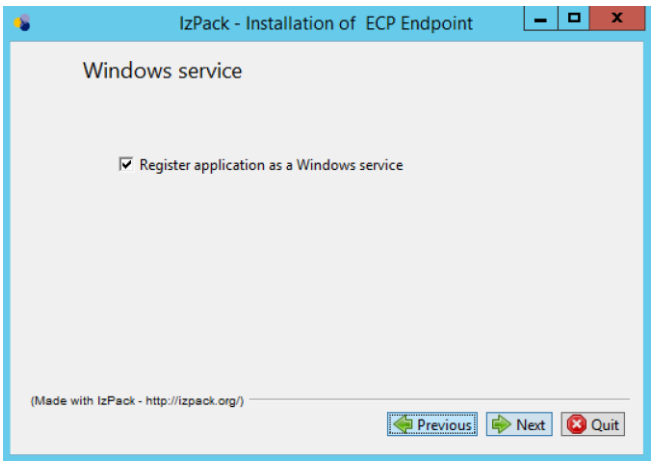

Figure 18: Installation 2

- ☐ *Click Next*
- ☐ *Choose whether the ECP Endpoint application should be automatically started after server start. The preferred option is to not start ECP Endpoint after server start.*

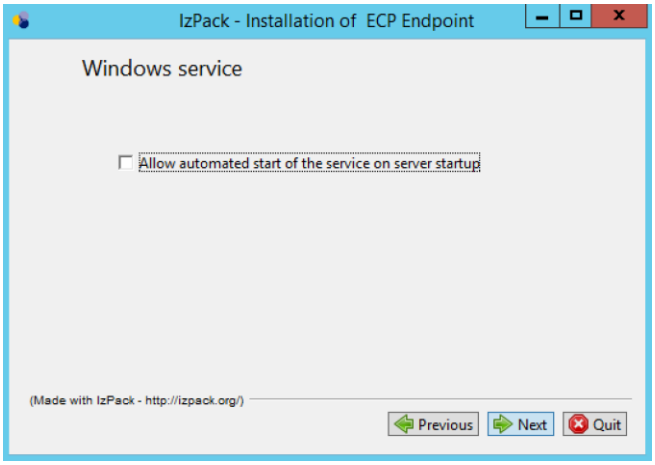

Figure 19: Installation 3

- ☐ *Click Next*
- ☐ *Application is installed*
- ☐ *Choose whether to use a default password ("**password**") in all configuration files and keystores or to define custom passwords.*

Figure 20: Installation – Password Selection

- ☐ *If custom passwords are selected, detailed option for custom settings will appear.*
    - *Plain – User can define own passwords; they will be stored in plain form in properties files*
    - *Hashed – User passwords are hashed by hashing function SHA-512*
    - *Encrypted – Passwords are encrypted by encryption algorithm (selected in the next step) with filled in key.*

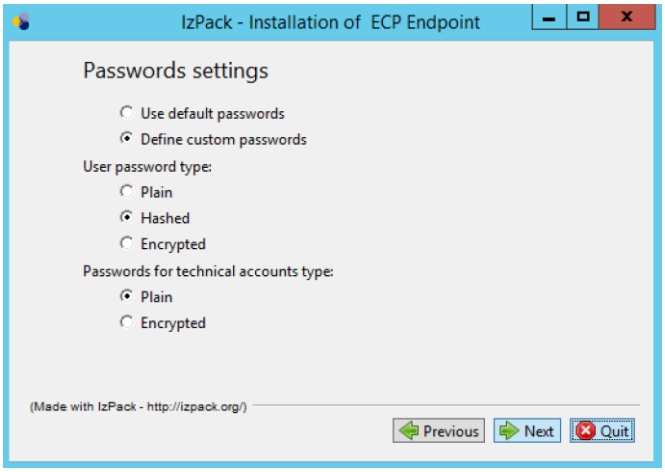

Figure 21: Installation – Password Settings

- ☐ *If the encrypted option is selected, encryption configuration parameters has to be filled in.*
    - *Encryption key – Key used for password encryption*
    - *Encryption algorithm*

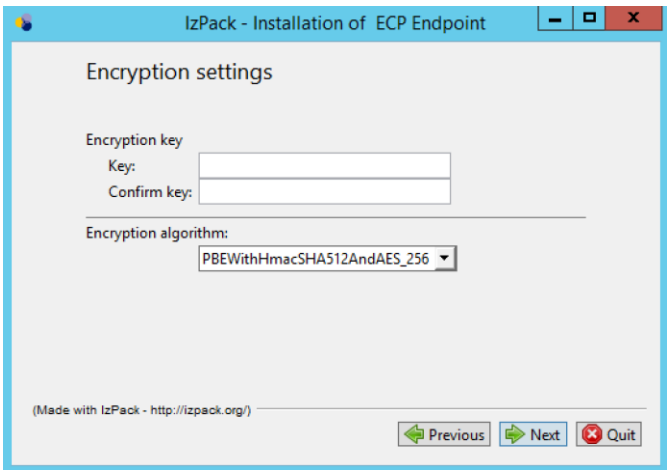

Figure 22: Installation – Encryption Settings

- ☐ *Click Next*
- ☐ *User password screen – type in administrator password*

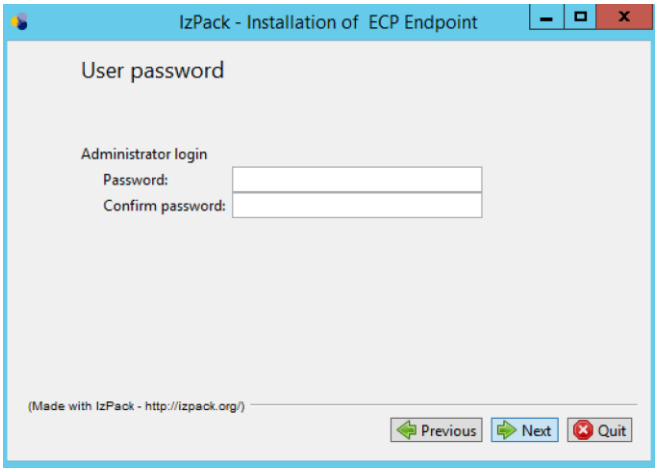

Figure 23: Installation – User Password

- ☐ *Click Next*
- ☐ *Technical accounts passwords*
  - *Authentication keystore – password to the authKeystore.jks*
  - *Internal Broker – password used for a connection to the Artemis Internal Broker*
  - *Keystores in database – password of keystore stored in database*
  - *CSRF secret – secret used in the creation of the CSRF token hash*

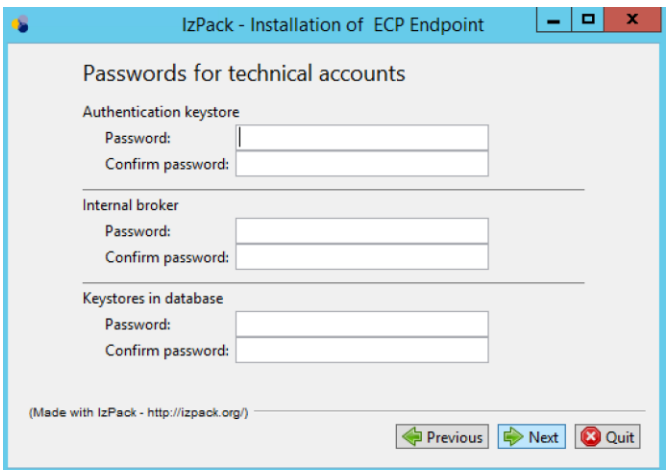

Figure 24: Installation – Technical Accounts 1

- ☐ *Click Next*
- ☐ *Custom passwords are configured*
- ☐ *If desired, an automatic installation script can be generated. It can be used in the future to install the application using the same options.*

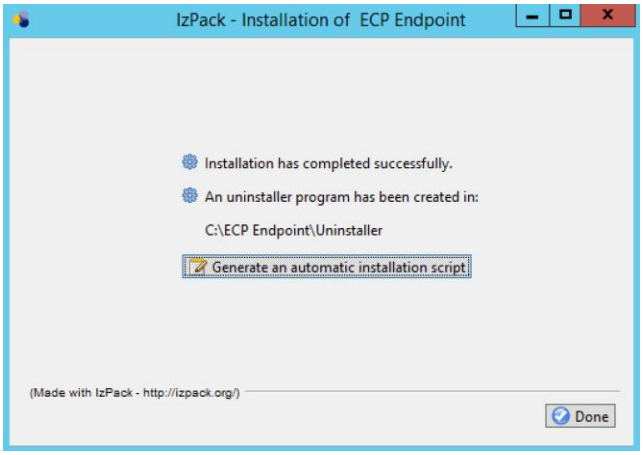

Figure 25: Installation 4

- ☐ *Click Done*

6. ☐ *Configure ECP Endpoint (see chapter ECP Endpoint configuration)*

8.2.3.2. Command line installation

Open command line with the administrator rights:

- ☐ *Start > type "cmd" > right click > run as administrator*
- ☐ *Change directory to location of <ecp-jar>, e.g.*
  - *cd c:\ecp-installer*
- ☐ *Run the ecp-endpoint.jar file using*
  - *java -jar <ecp-jar> -console*
- ☐ *Follow the console installation wizard*

Then configure ECP Endpoint before starting it (see ECP Endpoint configuration).

8.2.4. Starting and stopping ECP Endpoint

8.2.4.1. Using GUI

c) If installed as a service:

- *Start > Run > "services.msc"*
- *Find service with name like "ecp-endpoint" and description "ecp-endpoint"*
- *Start / stop service by buttons on the top left*

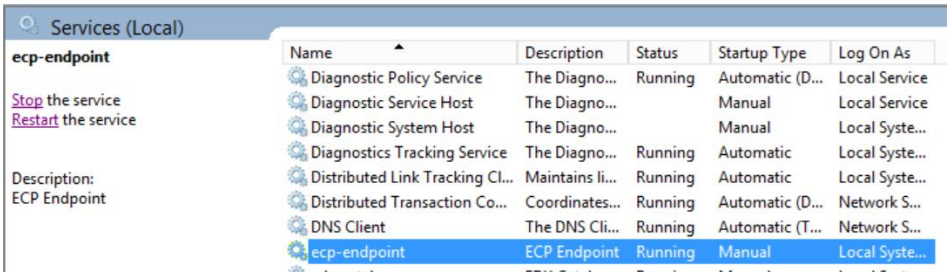

Figure 26: Services

d) If not installed as a service:

- *Start / stop application by using*
  - <install_path>\tomcat\bin\startup.bat
  - <install_path>\tomcat\bin\shutdown.bat

8.2.4.2. Using Command line

c) If installed as a service:

- *Start service using*
  - *SC start ecp-endpoint*
- *Stop service using*
  - *SC stop ecp-endpoint*

d) If not installed as a service:

- *Start application using*
  - *<install_path>\tomcat\bin\startup.bat*
- *Stop application using*
  - *<install_path>\tomcat\bin\shutdown.bat*

8.2.5. Installation verification

The application should be installed in the <install_path> folder. This installation folder should contain configuration files, Tomcat folder, and uninstaller. After the application is successfully started, it creates additional folders for data and log files. Default names and locations of these folders are *<install_path>\data* and *<install_path>\logs*.

8.2.5.1. Service status

The status of the service, if installed, can be checked either via command line:

- *SC query ecp-endpoint*

or in the Windows Services tool.

8.2.5.2. Log files

Check the application log files for more information about its status. The default location is *<install_path>\logs*.

8.2.5.3. GUI

When all previous steps were successfully done it should be possible to open GUI in the web browser.

If you use a default keystore for the HTTPS connector, it will be necessary to add a security exception in the browser, because ECP installation keystore does not contain Subject Alternative Name and it is self-signed.

To access GUI, open the following URL:

- *https://<endpoint-ip>:8443/ECP_MODULE*

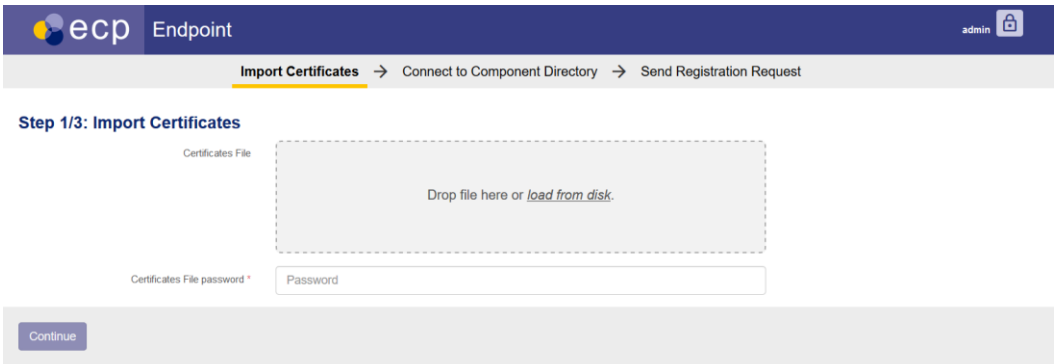

Figure 27: GUI – Verification

8.3. Uninstallation on Windows

Stop the application and run the command line as an administrator. Navigate to *<installation directory>\Uninstaller* and execute the following command:

- *java -jar uninstaller.jar -c*

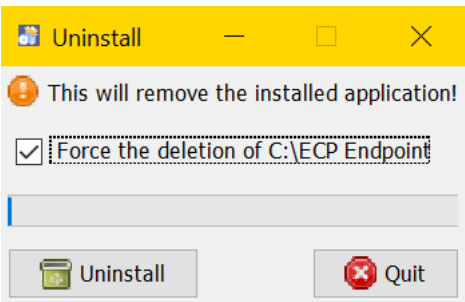

Figure 28: Uninstallation 1

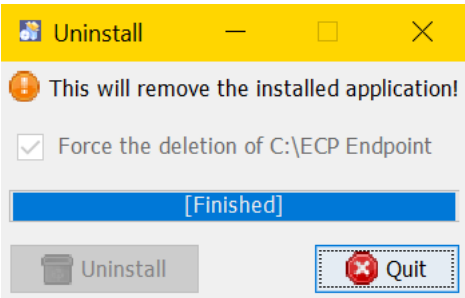

Figure 29: Uninstallation 2

Uninstaller will remove all application components and unregister the service if present.

8.4. Installation on Linux

The following section describes the installation of ECP Endpoint on the Linux platform.

8.4.1. Variables used in the guide

Several variables are used throughout this installation guide using the <variable_name> format. These variables are used only within this guide and are not to be confused with variables of your OS.

Name	Description
<jre-rpm>	Path to the JRE 17 installation package
<jre_install_path>	Location of the JRE 17 used by application
<ecp-rpm>	Path to the ECP Endpoint installation package
<install_path>	Path pointing to the target installation directory
<log_path>	Path to the ECP Endpoint log directory
<version>	Arbitrary version name

8.4.2. Prerequisites

Download the ECP Endpoint rpm file and copy it to the server.

Before processing, please check the *General Installation Prerequisites*.

Please note that ECP rpm file has dependency to specified by jre therefore installed Java must provide jre. Java headless OpenJDK distribution can be also used, however it will not pass the rpm dependency check. To overcome this issue, install rpm with nodeps option.

8.4.3. ECP application installation

8.4.3.1. Install using RPM package

Install ECP Endpoint. Change the directory to the folder where the ECP rpm installer is located and then invoke rpm install.

- ☐ *rpm -i <ecp-rpm>*

By default, ECP Endpoint is set up to automatically start after server start. The automatic start can be disabled by the following command:

- ☐ *systemctl disable ecp-endpoint*

Installer automatically creates the following folder structure:

Folder	Description
/usr/share/ecp-endpoint	Application binaries folder. This is the default location of <install_path>
/var/lib/ecp-endpoint	Application data folder
/var/log/ecp-endpoint	Application log folder. This is the default location of <log_path>
/etc/ecp-endpoint	ECP configuration files (**ecp.properties**)
/etc/systemd/system/ecp-endpoint.service.d/	Default configuration of ECP endpoint. Used to set JVM configuration and location of configuration files (**env.conf**).

8.4.4. Starting and stopping ECP Endpoint

After ECP Endpoint is configured, it can be started. ECP Endpoint is registered as a Linux service. The following commands are available:

- *To start ECP, use command "systemctl start ecp-endpoint.service"*
- *To stop ECP, use command "systemctl stop ecp-endpoint.service"*
- *To restart ECP, use command "systemctl restart ecp-endpoint.service"*

If not installed as a service:

- *Start application using bash script "<install_path>/tomcat/bin/startup.sh"*
- *Stop application using bash script "<install_path>/tomcat/bin/shutdown.sh"*

8.4.5. Installation verification

Check the status of the ecp-endpoint service:

- *systemctl status ecp-endpoint.service*
- *The service should be running*

Display application log (**ecp.log**) output:

- *tail <log_path>/ecp.log*

Display messages from a selected file:

- *tail <log_path>/catalina.out*
- *A similar line should appear in the **ecp.log** file: Started Endpoint in 25.001 seconds (JVM running for 31.924)*

Display the web interface of ECP

- *Open URL "https://<endpoint-ip>:8443/ECP_MODULE" in your web browser (replace the <endpoint-ip> with the correct IP address of your Endpoint server)*
- *If you use a default keystore for the HTTPS connector, it will be necessary to add a security exception in the browser, because ECP installation keystore does not contain Subject Alternative Name and it is self-signed.*
- *The page should display a registration page*

8.4.6. Log files

ECP Endpoint log files are available in the */var/log/ecp-endpoint* folder.

8.4.7. Application accounts

RPM ECP Endpoint installation automatically creates the user account *ecp-endpoint* with UID 2000 that belongs to the *ecp-endpoint* group with GID 2000. This account must be used for all file-based integration like fssf integration. Please note that it is not possible to change the UID and GID, so make sure that there is no other user using the same IDs.

To be able to login using ecp-endpoint user, it is necessary to set the password for user ecp-endpoint:

- *passwd ecp-endpoint*

8.4.8. Custom passwords

Configuration files and authentication keystore are delivered with default passwords. To define custom passwords, which will replace default passwords, run the *configure_passwords.sh* script in */etc/ecp-endpoint*.

After starting the script:

- ☐ *Select the desired form of user passwords*
  - *Plain – User can define own passwords; They will be stored in plain form in properties files*
  - *Hashed – User passwords are hashed by hashing function SHA-512*
  - *Encrypted – Passwords are encrypted by encryption algorithm (selected in the next step) with filled in key*
- ☐ *Select the desired form of technical passwords*
  - *Plain – User can define own passwords; They will be stored in plain form in properties files*
  - *Encrypted – Passwords are encrypted by encryption algorithm (selected in the next step) with filled in key*
- ☐ *If the encrypted passwords form has been selected for at least one of the password types, fill in the encryption key and the encryption algorithm (recommended is the one advised by the script)*
- ☐ *Fill all the custom passwords:*
  - *Administrator user password*
  - *Authentication keystore – password to the* ***authKeystore.jks***
  - *Internal Broker – password used for a connection to the Artemis Internal Broker*
  - *Keystores in database – password of keystore stored in database*
  - *CSRF secret – secret used in the creation of the CSRF token hash*

8.5. Uninstall on Linux

To uninstall ECP Endpoint, simply run the following command:

- ☐ *rpm -e ecp-endpoint*

8.6. Installation in High Availability mode

ECP Endpoint can be deployed in HA mode on two or more application servers. This setup consists of the following infrastructure components:

- *Two application servers with ECP Endpoint.*
- *Shared external database.*

It is recommended that a connected Internal Broker is also deployed in the High Availability mode, so there is no single point of failure.

8.6.1. Setup external database

Please see section *External Databases*.

8.6.2. Setup application servers

Please follow section *Installation on Windows* or *Installation on Linux* and install prerequisites and application on both application servers in the same way as for non-HA deployment.

8.6.3. Configure application

The application must be configured according to section *Parameters* required for HA installation.

8.6.4. Start and verify application

After installation, start only the first ECP Endpoint and go through the registration process (see section *ECP Endpoint Registration and Functional Smoke Test* for more details). When ECP Endpoint is successfully registered and synchronized, the remaining nodes may be started. Please follow section *Installation verification* (Windows) or *Installation verification* (Linux) to verify that the application has been installed correctly. It is recommended that each node be verified independently, one after the other. When finished, both nodes should be running.

8.7. ECP Endpoint configuration

Before starting ECP Endpoint, the application must be configured. The application is supplied with two configuration files.

Config file	Description
/etc/ecp-endpoint/ecp.properties	This is the main ECP configuration file with all the properties such as database connection, paths to filesystem, etc.
/etc/systemd/system/ecp-endpoint.service.d/env.conf	Configuration of JVM options (memory settings)

8.7.1. Configuration of ecp.properties

The **ecp.properties** configuration file comes with many configuration parameters. All the parameters are described in the Administration Guide. Most of the configuration parameters use default values, but the following parameters must be configured for HA installation:

8.7.1.1. Parameters required for HA installation

Parameter	Description
spring.profiles.active	To enable HA mode value must be set to *ecp-ha*. Please notice that *ecp-ha* profile can be also set to enable use of an external database when running component in a standalone mode. Profile *ecp-nonha* is intended for running component in standalone mode against embedded DB.
ecp.db.url	Connection URI to external database (If multi-line connection string is used, all lines, except the last one, must end with backslash). You may find the pattern appropriate for your database in section *External Databases*. Example: jdbc:mariadb://127.0.0.1:3306/ecp?characterEncoding=UTF-8&useSSL=false
ecp.db.username	Database user username. Example: ecp
ecp.db.password	Database user password. Example: password
spring.datasource.validation-query	This value should be set to a valid query for the selected database (see the *External Databases* chapter). Example for the embedded DB: 'select 1' (without quotation marks)
spring.datasource.driver-class-name	Driver for specific external database. For more information, see the *External Databases* chapter.
ecp.cluster.manager.localInstanceID	Node identifier in the cluster. This value must be unique. Example: E1a or server's hostname

8.7.2. Internal Broker configuration

By default, Endpoint will try to connect to Internal Broker in the localhost on 5672 port, and using a default Internal Broker public certificate, which is included in the Endpoint authKeystore.jks. It is necessary to change the configuration when Internal Broker is in a different server, does not use a default port, or it does not use a default keypair.

When Internal Broker does not use a default keypair, its public certificate must be imported to the Endpoint authKeystore.jks.

Parameter	Description	Default value
internalBroker.host	Internal Broker bind address.	127.0.0.1
internalBroker.ampq.port	Internal Broker bind port.	5672
internalBroker.parameters	Addition connection URL parameters	-
internalBroker.urls	Failover connection to Internal Broker. When configured, properties host, amqp.port and parameters are ignored. Example: failover:(amqp://<node1_IP>:5672?parameters, amqp://<node2_IP>:5672)	host + amqp.port + parameters
internalBroker.keystore.location	Path to the KeyStore with SSL certificate.	authKeystore.jks path
internalBroker.keystore.password	SSL keystore password.	Configured during the installation
internalBroker.auth.user	Sets Endpoint username. This user must exist in Internal Broker.	endpoint
internalBroker.auth.password	Endpoint user password for Internal Broker.	Configured during the installation
internalBroker.queuePrefix	A unique prefix, which will be added to every queue name belonging to Endpoint. Required when multiple Endpoints connect to one Internal Broker. Allowed characters are A-Z, a-z, digits 0-9, dot ., underscore _, and hyphen -.	-

8.7.3. TLSv1.2 support

By default, ECP Endpoint supports only TLSv1.3 protocol for incoming connections. If there is a Business Application which cannot use TLSv1.3, then TLSv1.2 must be manually enabled in the configuration files.

If the Business Application connects to the Endpoint HTTPS protocol, open ***server.xml*** file in the following path:

- *Windows:*
  - *<install_path>\tomcat\conf\server.xml*
- *Linux:*
  - */usr-share/ecp-endpoint/conf/server.xml*

- ☐ *Set "TLSv1.2,TLSv1.3" in the sslEnabledProtocols attribute*
- ☐ *Set these ciphers in ciphers attribute:*
  - *TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256:TLS_AES_128_GCM_SHA256:ECDHE-RSA-AES256-GCM-SHA384:ECDHE-RSA-AES128-GCM-SHA256:DHE-RSA-AES256-GCM-SHA384:DHE-RSA-AES128-GCM-SHA256*

Refer to the *Internal Broker configuration – TLSv1.2 support* chapter if the Business Application connects to the Endpoint by using AMQPS protocol.

9. ECP Endpoint Registration and Functional Smoke Test

9.1. Prerequisites

Before starting the Endpoint registration, please ensure that the following criteria are met:

- *The Component Directory administrator has provided you with the Component Directory URL, Component Directory code, and registration certificate with password (if not, please apply for it with your EIC X code and EIC V code).*
- *Configured Internal Broker is running*
- *ECP Endpoint is configured correctly, and service is started*

9.2. Registration process

- ☐ *Open a web browser and navigate to the URL of your Endpoint (e.g. https://<endpoint-ip>:8443/ECP_MODULE).*
- ☐ *If you use a default keystore for the HTTPS connector, it will be necessary to add a security exception in the browser, because ECP authentication keystore does not contain Subject Alternative Name*

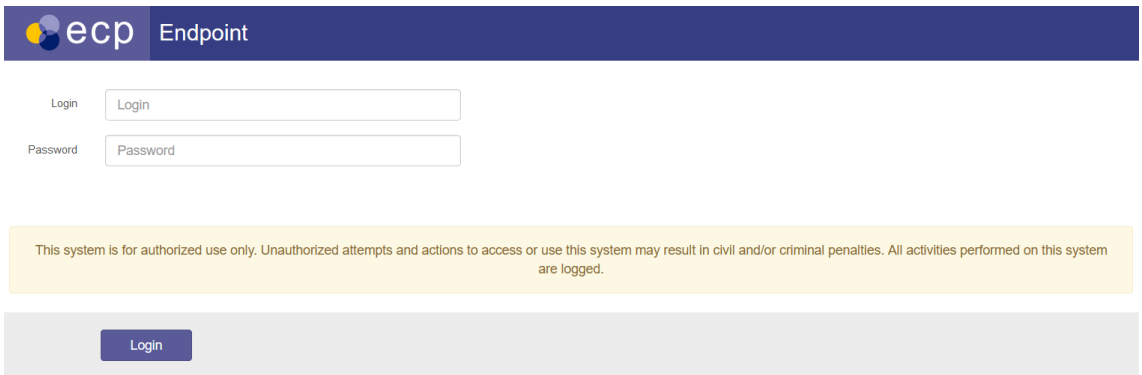

Figure 30: Login

- ☐ *Login with a user and a password. Depending on the configured authentication method, the user must exist in the ecp-users.properties configuration file or LDAP server.*

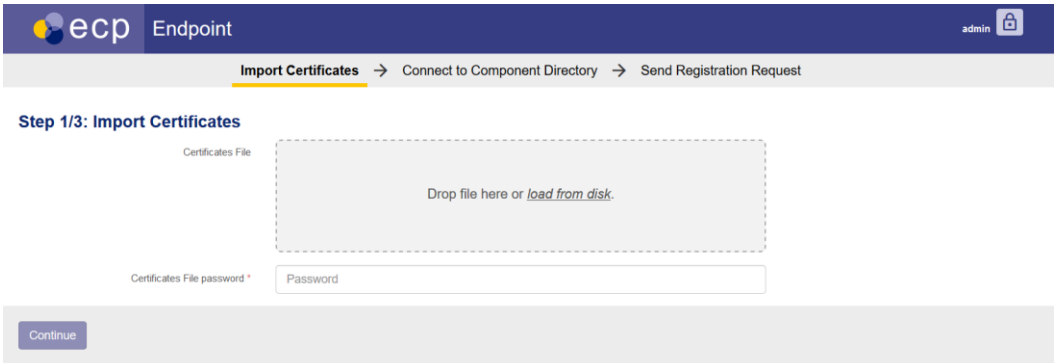

Figure 31: Registration 1

- ☐ *Insert the registration keystore provided by your Component Directory administrator and adjusted password, wait for the file to upload. Click "Continue" to proceed to the next step.*

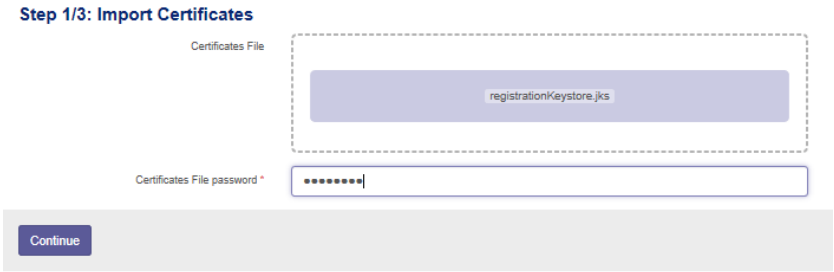

Figure 32: Registration 2

- ☐ *Enter the Component Directory code and URL (e.g. https://<directory-ip>:8443/ECP_MODULE) provided by the Component Directory administrator and then click on the check connectivity. Please note that a successful connection check is required to proceed to the next step.*
- ☐ *When NATing is enabled in Endpoint, Component Directory URL is translated when there is a match in the configured NAT table*

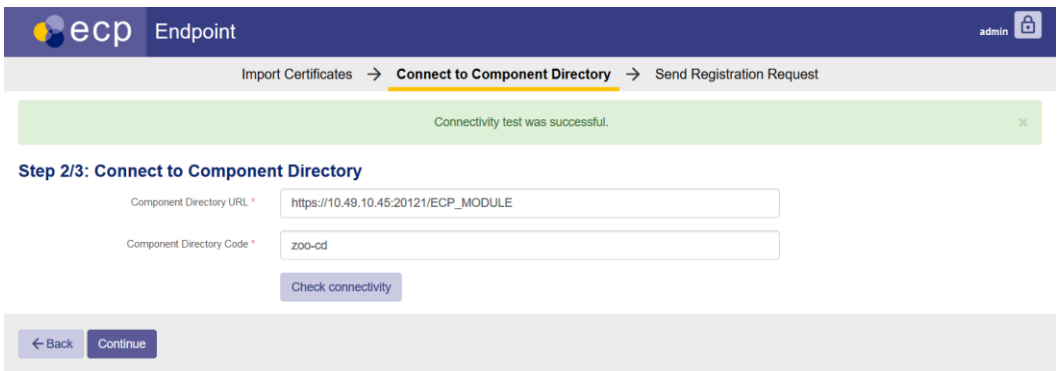

Figure 33: Registration 3

- ☐ *Fill in your component v-code (allowed characters are A-Z, a-z, digits 0-9, at-sign @ and hyphen -).*
- ☐ *Optionally fill in Project and Environment names, which will be displayed in the ECP header (allowed characters are A-Z, a-z, digits 0-9 and hyphen -, maximum length is 16 characters).*
- ☐ *Fill in your company information (please note that the phone number must be in international format - e.g. 00<country code><number>).*
- ☐ *Accept the GDPR disclaimer.*
- ☐ *Click "Submit request".*
- ☐ *If the registration keystore is provided specifically for your component, the component code is pre-filled and cannot be changed.*

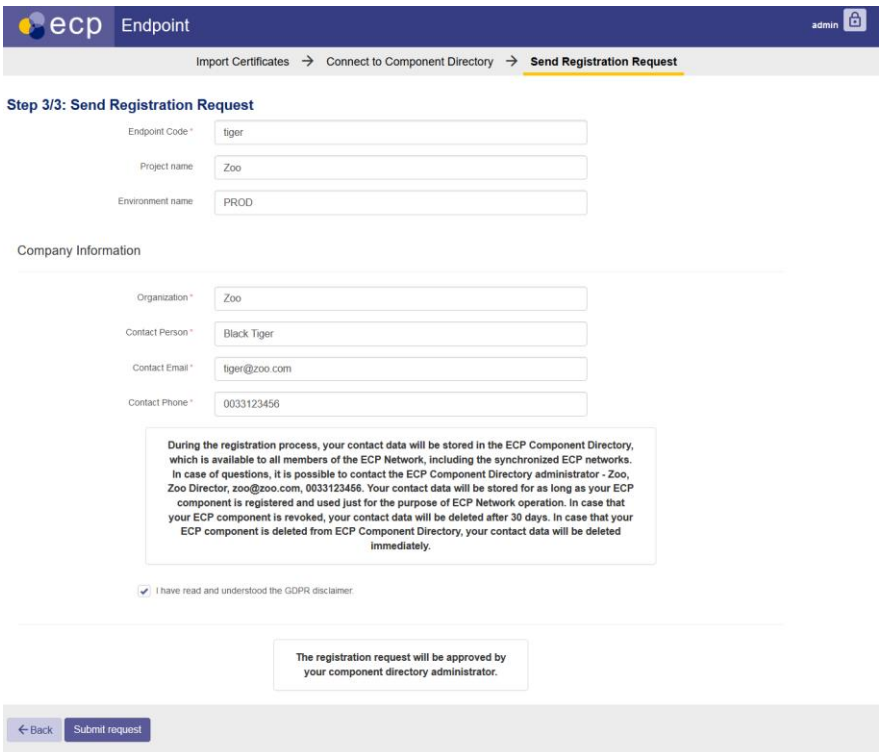

Figure 34: Registration 4

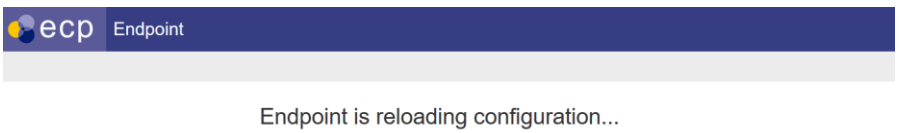

Figure 35: Registration 5

- ☐ *At this point, you must wait until the Component Directory administrator approves your registration request. Once the request is approved, your Endpoint automatically downloads the results and becomes operational.*

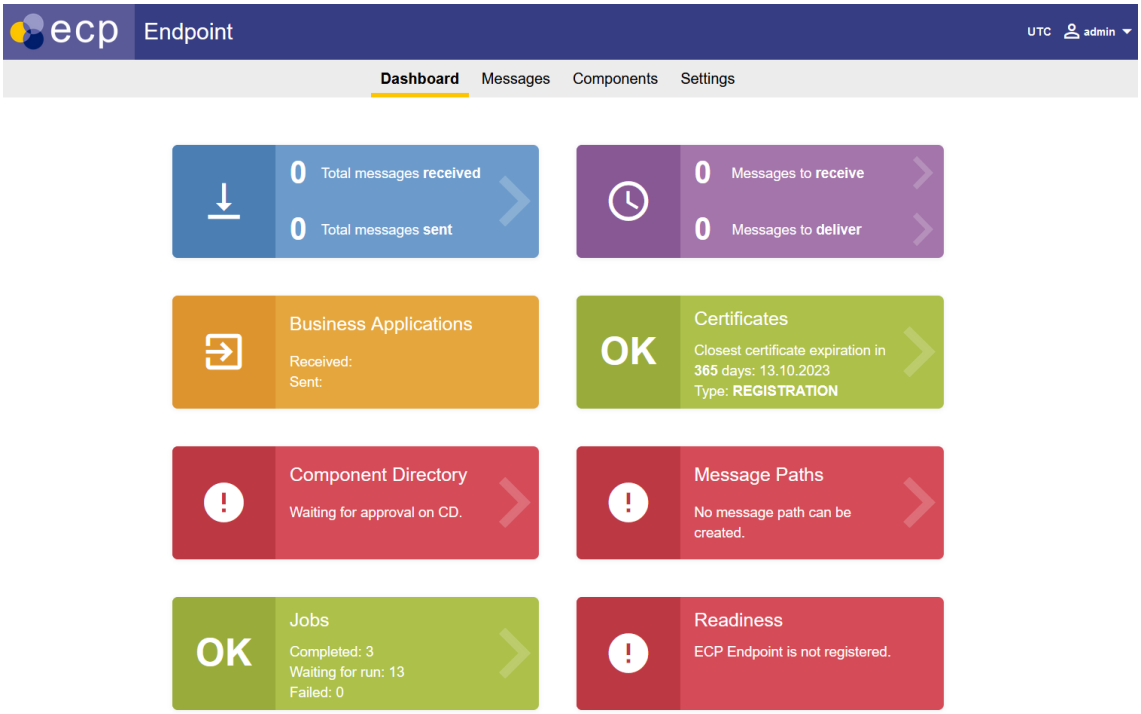

Figure 36: Dashboard

9.2.1. Restore registration process

It is possible to run registration process again, for example if there was some input mistake during the first run. In *Settings* page, click on *Initiate Registration* button and make confirmation in a modal window. You will be then redirected to the first page of the registration process, and you can proceed with a registration.

Warning: All ECP Endpoint data will be removed from a database and keystores when a new registration process is confirmed.

9.3. AMQP Context

This chapter serves only as an example. It is not necessary to configure the AMQP channel when it is not required. Please refer to the *Public Interface* document for descriptions of other integration channels.

- *To be able to send messages and test their reception, there is a need to configure the AMQP communication channel. This will allow the messages to get through to the ECP Broker and other ECP Endpoints.*

- *In the GUI, go to Settings – Runtime Configuration, click on the "Edit Configuration" button.*
- *Add these values to the following parameters:*

  ecp.endpoint.amqpApiEnabled = true

  ecp.endpoint.sendHandler[0].beanName = amqpApiSendHandler

  ecp.endpoint.sendHandler[0].typeName = *
- *Click on the Save button*

9.4. Functional Smoke Test

It is possible to go through following steps on GUI to test your component functionality.

9.4.1. Configure message path

- *Go to Settings – Paths and click on "+ New Path" button.*
- *Fill in the form that appeared. It is recommended to set Senders to All and Message Type to * (anything). You need to choose Indirect and then select a Broker which is available in your network (if there is no possibility, please contact your administrator). Click on save.*

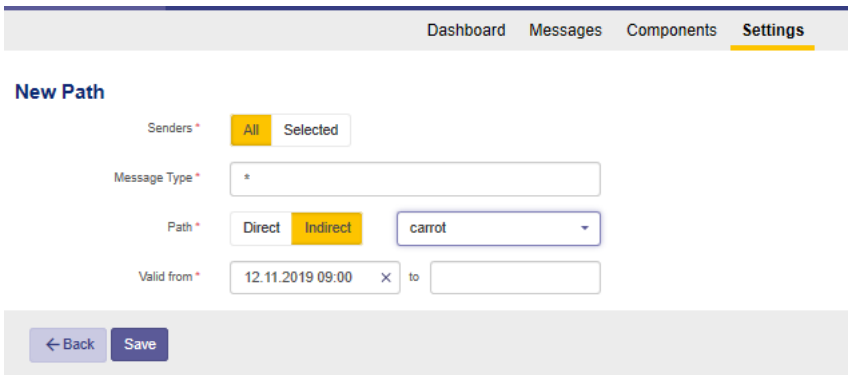

Figure 37: Message Path

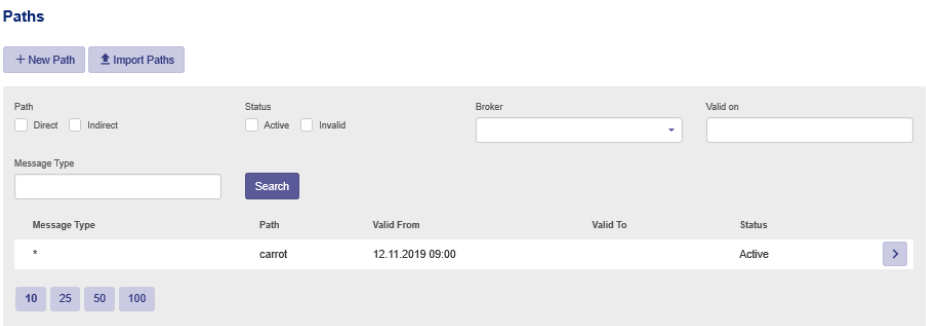

Figure 38: Message Path 2

9.4.2. Send Message

- *Please check that the AMQP configuration was successfully done before this step.*
- *Go to Messages and click on "New Message" button. Fill in the form that appeared. It is recommended to send message to itself – use your v-code for receiver. Send message.*

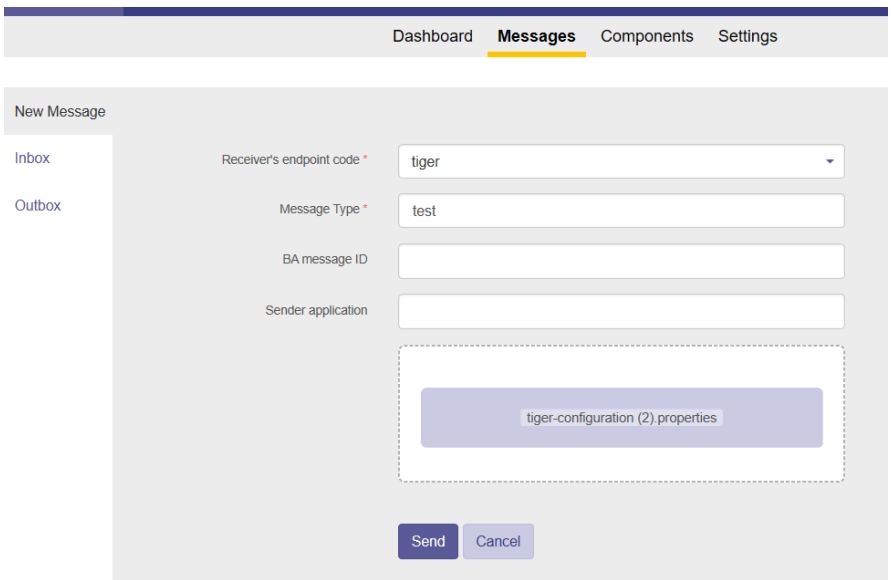

Figure 39: New message

- *If everything is configured correctly, your message should have the state Received. Please check both Inbox and Outbox.*

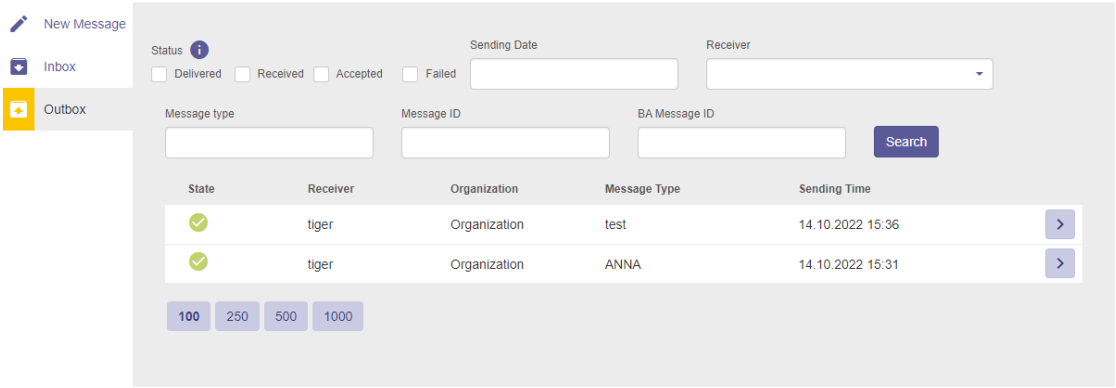

Figure 40: Send Message

- *Functional Smoke Test is successfully done.*

10. ECP Component Directory Installation

When High Availability mode is required, first proceed with *Installation* chapter, then proceed with *Installation in High Availability mode* chapter.

10.1. Common Prerequisites

10.1.1. General prerequisites before installing ECP Component Directory

- *ECP Component Directory rpm package*
- *Registration keystore (this keystore will be provided by the ECP network administrator)*
- *Component Directory code (this should be defined or agreed to with the ECP network administrator)*
- *Before processing, please check the General Installation Prerequisites.*

10.2. Installation

The following section describes the installation of the ECP Component Directory.

10.2.1. Variables used in the guide

Several variables are used throughout this installation guide using the <variable_name> format.

Name	Description
<jre-rpm>	Path to the JRE 17 installation package
<jre_install_path>	Location of the JRE 17 used by application
<ecp-rpm>	Path to the ECP Component Directory installation package
<install_path>	Path pointing to the target installation directory

<log_path>	Path to the ECP Component Directory log directory
<version>	Arbitrary version name

10.2.2. Prerequisites

10.2.2.1. ECP Component Directory installer

Download the ECP Component Directory rpm file and copy it to the server.

10.2.2.2. JRE

Before processing, please check the *General Installation Prerequisites*.

10.2.3. ECP Component Directory application installation

10.2.3.1. Install using RPM package

Install the ECP Component Directory. Change directory to the folder where the ECP RPM is located and invoke rpm install.

- ☐ *rpm -i <ecp-rpm>*

By default, ECP Component Directory is set up to automatically start after server start. The automatic start can be disabled by the following command:

- ☐ *systemctl disable ecp-directory*

Installer automatically creates the following folder structure:

Folder	Description
/usr/share/ecp-directory	Application binaries folder. This is the default location of <install_path>
/var/lib/ecp-directory	Application data folder
/var/log/ecp-directory	Application log folder. This is the default location of <log_path>
/etc/ecp-directory	ECP Component Directory configuration files
/etc/systemd/system/ecp-directory.service.d/	Default configuration of ECP Component Directory. Used to set JVM configuration and location of configuration files. (file /etc/systemd/system/ecp-directory.service.d/env,conf)

10.2.4. Starting and stopping the ECP Component Directory

After the ECP Component Directory is configured (please see section *ECP Component Directory configuration*), it can be started. The ECP Component Directory is registered as a Linux service. The following commands are available:

- *To start ECP, use command "systemctl start ecp-directory.service"*
- *To stop ECP, use command "systemctl stop ecp-directory.service"*
- *To restart ECP, use command "systemctl restart ecp-directory.service"*

10.2.5. Installation verification

Check the status of the ecp-directory service:

- *systemctl status ecp-directory.service*
- *The service should be running.*

Display the application log (**ecp.log**) output:

- *tail <log_path>/ecp.log*

Display messages from a selected file:

- *tail <log_path>/catalina.out*
- *A similar line should appear in the ecp.log file: Started Component Directory in 17.785 seconds (JVM running for 25.084)*

Display the web interface of ECP

- *Open URL "https://<directory-ip>:8443/ECP_MODULE" in your web browser (replace the <directory-ip> with the correct IP address of your Component Directory server)*
- *If you use a default keystore for the HTTPS connector, it will be necessary to add a security exception in the browser, because ECP installation keystore does not contain Subject Alternative Name and it is self-signed.*
- *The page should display a registration page.*

10.2.6. Application accounts

The ECP Component Directory automatically creates a user account *ecp-directory* with UID 2000 that belongs to the *ecp-directory* group with GID 2000. Please note that it is not possible to change the UID and GID, so make sure that there is no other user using the same IDs.

10.2.7. Custom passwords

Configuration files and authentication keystore are delivered with default passwords. To define custom passwords, which will replace default passwords, run the *configure_passwords.sh* script in */etc/ecp-directory*.

After starting the script:

- ☐ *Select the desired form of user passwords*
    - *Plain – User can define own passwords; they will be stored in plain form in properties files*
    - *Hashed – User passwords are hashed by hashing function SHA-512*
    - *Encrypted – Passwords are encrypted by encryption algorithm (selected in the next step) with filled in key*
- ☐ *Select the desired form of technical passwords*
    - *Plain – User can define own passwords; they will be stored in plain form in properties files*
    - *Encrypted – Passwords are encrypted by encryption algorithm (selected in the next step) with filled in key*
- ☐ *If the encrypted passwords form has been selected for at least one of the password types, fill in encryption key and encryption algorithm (recommended is the one advised by the script)*
- ☐ *Fill all the custom passwords:*
    - *Administrator user password*
    - *Authentication keystore – password to the* ***authKeystore.jks***
    - *Registration keystore – password to the* ***registrationKeystore.jks****, which is downloaded for registration of other components*
    - *CA keystore – password for the ECP CD CA certificate*
    - *Keystores in database – password of keystore stored in database*
    - *CSRF secret – secret used in the creation of the CSRF token hash*

10.3. Uninstall

To uninstall ECP Component Directory, simply run the following command:

- ☐ *rpm -e ecp-directory*

10.4. Installation in High Availability mode

The ECP Component Directory can be deployed in HA mode on two or more application servers. This setup consists of the following infrastructure components:

- *Two or more application servers with ECP Component Directory.*
- *Shared external database.*

10.4.1. Setup application servers

Please follow section *Installation* and install prerequisites and application on both application servers.

10.4.2. Configure application

Application must be configured according to section *Parameters required for HA installation*

.

10.4.3. Start and verify application

After installation, start only the first ECP Component Directory and go through the registration process (see chapter *ECP Component Directory Registration* for more details). When the ECP Component Directory is successfully registered and synchronized, the remaining nodes may be started. For the initial start-up of other nodes, it is important to let the node start and then restart it. This procedure is required for Tomcat to use correct SSL certificates. Please follow section *Installation verification* to verify that the application has been installed correctly. It is recommended that each node be verified independently, one after the other. When finished, both nodes should be running.

10.5. ECP Component Directory configuration

Before starting the ECP Component Directory, the application must be configured. The application is supplied with two configuration files.

Config file	Description
/etc/ecp-directory/ecp-directory.properties	This is the main ECP configuration file with all the properties such as database connection etc.

10.5.1. Configuration of ecp-directory.properties

The ***ecp-directory.properties*** configuration file comes with many configuration parameters (all of them are described briefly in the ***ecp-directory.properties***). Most of the configuration parameters use default values, but the following parameters must be configured for HA installation:

10.5.1.1. Parameters required for HA installation

Parameter	Description
spring.profiles.active	To enable HA mode value must be set to *ecp-ha*. Please notice that *ecp-ha* profile can be also set to enable use of an external database when running component in standalone mode. Profile *ecp-nonha* is intended for running component in standalone mode against embedded DB.
ecp.db.url	Connection URI to external database (If multi-line connection string is used, all lines, except the last one, must end with backslash). Example: jdbc:mariadb://127.0.0.1:3306/ecp-directory?characterEncoding=UTF-8&useSSL=false
ecp.db.username	Database user username. Example: ecp-directory
ecp.db.password	Database user password. Example: password
spring.datasource.validation-query	This value should be set to a valid query for the selected database (see the *External Databases* chapter). Example for the embedded DB: ‘select 1’ (without quotation marks)

spring.datasource.driver-class-name	Driver for specific external database. For more information, see the *11. External Databases* chapter.
ecp.cluster.manager.localInstanceID	Node identifier in the cluster. This value must be unique. Example: server's hostname

Other contents of these files are described in the Administration Guide.

11. ECP Component Directory Registration

11.1. Prerequisites

Before starting the Component Directory registration, please ensure that following criteria are met:

- *Component Directory is running*

11.2. Registration process

- ☐ *Open a web browser and navigate to the URL of the ECP Component Directory (e.g. https://<directory-ip>:8443/ECP_MODULE)*
- ☐ *If you use a default keystore for the HTTPS connector, it will be necessary to add a security exception in the browser, because ECP authentication keystore does not contain Subject Alternative Name*
- ☐ *Login into the application. The default credentials are admin@password and can be changed in the configuration.*
- ☐ *Insert the registration key store provided by the ECP network administrator, wait for the file to upload, and insert its password. Click "Next" to proceed to the next step.*
- ☐ *Fill in the component code (allowed characters are A-Z, a-z, digits 0-9, at-sign @ and hyphen -) and URL. The component code should be defined or agreed to with the ECP network administrator. The URL is https URL for your ECP Component Directory (e.g. https://<directory-ip>:8443/ECP_MODULE).*
- ☐ *Optionally fill in Project and Environment names, which will be displayed in the ECP header (allowed characters are A-Z, a-z, digits 0-9 and hyphen -, maximum length is 16 characters).*
- ☐ *Fill in your company information (please note that the phone number must be in international format - e.g. 00<country code><number>).*
- ☐ *Accept the GDPR disclaimer.*
- ☐ *Click "Submit request".*

12. ECP Broker Installation

The Broker application is installed using the installation RPM package provided.

When High Availability mode is required, first proceed with this chapter, then proceed with *High available installation* chapter.

12.1.1. Variables used in the guide

Name	Description
<jre_install_path>	Location of the JRE 17 used by application
<version>	Arbitrary application version name
<ecp-rpm>	Path to the ECP Broker installation package

12.1.2. Prerequisites

12.1.2.1. ECP Broker installation RPM package

Download the ECP Broker installation file *<ecp-rpm>* and copy it to the server.

12.1.2.2. Registration keystore

The registration keystore will be provided by your ECP Component Directory administrator.

12.1.2.3. JRE

Before processing, please check the *General Installation Prerequisites*.

12.1.2.4. Firewall

Before processing, please check the *General Installation Prerequisites*.

12.1.3. Installation

Install the ECP Broker. Change directory to the folder where the ECP Broker RPM is located and invoke rpm install.

- ☐ *rpm -i <ecp-rpm>*

By default, ECP Broker is set up to automatically start after server start. The automatic start can be disabled by the following command:

- ☐ *systemctl disable ecp-broker*

Installer automatically creates the following folder structure:

Folder	Description
/opt/ecp-broker/artemis	Folders for Artemis core files
/opt/ecp-broker/broker	Folders for Artemis broker instance
/opt/ecp-broker/broker/etc	Artemis broker instance configuration files
/opt/ecp-broker/broker/data	Artemis broker instance data
/opt/ecp-broker/cd	ECP Component Directory content storage
/opt/ecp-broker/config	ECP Broker configuration files
/opt/ecp-broker/reg	ECP Broker registration tool

- ☐ *Update /opt/ecp-broker/config configuration settings. Follow the instructions in the ECP Broker configuration chapter.*
- ☐ *Use the Broker registration tool to register Broker with the ECP Component Directory. Follow the instructions in section ECP Broker Registration.*

12.1.4. Starting and stopping ECP Broker

After the ECP Broker is configured and registered, it can be started. The following commands are available:

- *Start the Broker: "systemctl start ecp-broker.service"*
- *Stop the Broker: "systemctl stop ecp-broker.service"*
- *Restart the Broker: "systemctl restart ecp-broker.service"*

12.1.5. Installation verification

The application should be installed in the */opt/ecp-broker* folder. This installation folder should contain the previously described folder structure.

12.1.5.1. Broker status

The status of the Broker can be checked via command line:

- *systemctl status ecp-broker.service*

12.1.5.2. Log files

Check the application log files for more information about its status. The default location is /opt/ecp-broker/broker/log.

12.1.5.3. Artemis GUI

GUI of Artemis Broker is started on port 8161. By default, the access is limited for opening it on the server where the Artemis is installed.

For allowing connection to GUI from other locations it is necessary to change the configuration file */opt/ecp-broker/broker/etc/**jolokia-access.xml***.

> *Add <allow-origin> element with an URL pattern to match URLs from where the access will be allowed*

> *Multiple <allow-origin> elements can be defined*

> *URL pattern supports a wildcard character: **

Also, it is necessary to specify the URL in the */opt/ecp-broker/broker/etc/**bootstrap.xml***:

> *In the binding element, find and change the uri attribute:*

- *E.g. uri="http://0.0.0.0:8161"*

  or

- *uri="http://<server IP address>:8161"*

Please note that the Artemis Prometheus metrics plugin, which exposes the metrics at **/metrics* URL, does not support the authentication. If you wish to enable a remote access to the console without exposing the metrics, it is possible to disable the metrics in */opt/ecp-broker/broker/etc/**bootstrap.xml***:

> *In the binding element, find and remove this line:*

- *<app url="metrics" war="metrics.war"/>*

12.1.6. Application accounts

The ECP Broker automatically creates a user account *ecp-broker* with UID 2000 that belongs to the *ecp-broker* group with GID 2000. Please note that it is not possible to change the UID and GID, so make sure that there is no other user using the same IDs.

When ECP Component Directory is already installed on the same server, UID 2001 and GID 2001 is used instead.

12.1.7. Custom passwords

The installation comes with one preconfigured user with a default password. The default user is used only with a standard authentication, it does not work when the LDAP authentication is enabled.

To define a custom password, which will replace the default password, run the *configure_passwords.sh* script in */opt/ecp-broker/broker/etc*.

After starting the script:

- ☐ *Select the desired form of the user password*
    - *Plain – Password will be stored in a plain form in the configuration file*
    - *Hashed – Password will be hashed*
- ☐ *Fill the custom user password:*

The script can change only the default password. For a next change, or when additional users are added, it is necessary to define passwords manually in the configuration file */opt/ecp-broker/broker/etc/artemis-users.properties.*

Artemis provides a command to create a hash from an input, which can be then used in *artemis-users.properties:*

> */opt/ecp-broker/broker/bin/artemis mask --hash "<password>"*

Hashed password must be wrapped in ENC() in the properties file.

12.2. Uninstall

To uninstall ECP Broker, simply run the following command:

- ☐ *rpm -e ecp-broker*

12.3. High available installation

The ECP Broker can be deployed in HA mode on two or more application servers. This setup consists of the following infrastructure components:

- *Two or more application servers with ECP Broker.*
- *Shared file storage.*

If you wish to use an external database instead of the shared file storage, please refer to the Artemis documentation.

12.3.1. Setup application servers

Please follow the installation procedure. Only one server is needed for initial setup.

12.3.2. Configure application

Open ***broker.xml*** in */opt/ecp-broker/broker/etc* folder.

Configure a path to the shared storage, which is accessible by all nodes, in these elements:

- *<paging-directory>*
- *<bindings-directory>*
- *<journal-directory>*
- *<large-messages-directory>*

Find and uncomment the *<connectors>* element.

- *Inside the element, keep only one connector with a name according to the node type: primary (master), or backup (slave)*
- *Configure the connector IP and port. It must be accessible from other nodes.*

Uncomment the *"artemis"* acceptor in the *<acceptors>* element.

- *Configure the acceptor port to match the port in the connector.*

Find and uncomment the *<cluster-user>* and *<cluster-password>* elements.

- *Change the default values. User and password must be the same in all nodes.*

Find and uncomment the *<cluster-connections>* element.

- *Inside the element, keep only one connector-ref and static-connectors according to the node type:*
    - *Primary (master)*

        <connector-ref>master</connector-ref>

<static-connectors>

<connector-ref>slave</connector-ref>

</static-connectors>

- *Backup (slave)*

<connector-ref>slave</connector-ref>

<static-connectors>

<connector-ref>master</connector-ref>

</static-connectors>

Find and uncomment the *<ha-policy>* element.

- *Inside the element, keep only <master> or <slave> element according to the node type.*

For more details about the Artemis configuration, please refer to the official Artemis documentation. ECP Broker does not alter the Artemis High Availability behaviour, so any Artemis configuration will work with ECP Broker.

12.3.3. Start and verify application, install remaining nodes

After installation, start one node with the ECP Broker and go through the registration process (see section *ECP Broker Registration* for more details). When the ECP Broker is successfully registered and synchronized, the remaining nodes may be set up.

To set up the remaining nodes, stop the Broker, copy the current installation directory, and distribute it to other servers. In **broker.xml**, any configuration referencing the primary (master) node must be changed to the backup (slave) node.

12.4. ECP Broker configuration

Before starting the ECP Broker, the application must be configured and registered by Component Directory. The application comes with many configuration files:

- ***broker.properties*** *- This is the configuration file with all ECP-related properties like component information or Component Directory synchronization, used by the ECP plugin. File location is /opt/ecp-broker/config/broker.properties*
- ***Artemis configuration files*** *in /opt/ecp-broker/broker/etc. These are configuration files used by the Artemis broker, for example for the authentication, logging, or memory limits.*

12.4.1. Configuration of broker.properites

The configuration file ***broker.properties*** comes with configuration parameters which are briefly described in the ***broker.properties*** file and also in this section). Some of the configuration parameters use a default value, but the following parameters must be configured for each installation:

12.4.1.1. Registration process properties

Parameter	Description
ecp.broker.contactPerson	Broker registration data
ecp.broker.contactEmail	Broker registration data
ecp.broker.contactPhone	Broker registration data
ecp.broker.organization	Broker registration data
ecp.broker.urls	Broker URL with pattern <protocol>://<host>:<port>. Broker protocol is typically amqps. Multiple values are separated by comma (for HA deployment). e.g. amqps://192.168.1.1:5671
ecp.broker.networks	Comma separated list of physical network names to which the URLs (**ecp.broker.urls**) belong. The number of networks must be equal to the number of URLs and the order must match because URL and network forms a pair. e.g. Internet
ecp.broker.code	ECP Broker component code
ecp.broker.filter.components	Define which components can communicate through Broker. Multiple values of component codes separated by comma. Wildcard character * can be used to allow communication for all components. Default value is *
ecp.broker.filter.types	Define which message types are accepted by the Broker. Multiple values are separated by comma Wildcard character * can be used to accept all message types. Default value is *

12.4.1.2. Broker Component Directory properties

Parameter	Description

ecp.directory.client.synchronization.homeComponentDirectoryPrimaryCode	Component code of home Component Directory
ecp.directory.client.synchronization.homeComponentDirectoryPrimaryUrl	URL of home Component Directory. CD URL with the following pattern: <protocol>://<host>:<port>/ECP_MODULE. CD protocol is typically https. (E.g. *https://127.0.0.1:8443/ECP_MODULE*)
ecp.directory.client.root	Customizable location of Component Directory data stored on Broker. Location must exist at Broker filesystem. e.g.: /opt/ecp-broker/cd

12.4.1.3. Broker keystores properties

Parameter	Description
ecp.keystore.location	Location of a keystore with registration certificates. This keystore is provided by the Component Directory administrator and was listed in the prerequisites. After the Broker registration, an authentication keypair is added to this keystore. This keypair is also updated when the certificate renewal was made.
ecp.keystore.password	Registration keystore password (by default this is “password”)
ecp.authKeystore.location	Location of the keystore with authentication certificates (e.g. /opt/ecp-broker/authKeystore.jks). This location must point to an unexisting file as it will be used by ECP to save a new keystore after a successful registration.
ecp.authKeystore.password	Authentication keystore password (e.g. password).

Other contents of these files are described in the Administration Guide.

12.4.2. Journal buffer timeout

Artemis maintains an internal buffer for the file journal instead of writing directly to the journal with every change. This buffer is flushed when it is full, or when a timeout expires.

Artemis provides a script for detecting an optimal timeout value:

> */opt/ecp-broker/broker/bin/artemis perf-journal*

Take the returned value and replace a default value in the *journal-buffer-timeout* element in */opt/ecp-broker/broker/etc/broker.xml* file.

13. ECP Broker Registration

13.1. Prerequisites

Before starting the registration of the Broker please ensure that following criteria are met:

- *Broker is configured (see section ECP Broker configuration)*
- *The Component Directory administrator has provided you with the Component Directory URL, Component Directory code, and registration certificate*
- *Your home Component Directory is running*

13.2. Registration process

The Broker installation package comes with a Broker registration tool (located in */opt/ecp-broker/reg*). This tool is used for the Broker registration and the Broker certificates update. The registration process is divided into three steps, all the steps must be executed successfully as specified to finish the registration process:

1. *Create a Broker registration request – creates a new Broker registration request and sends it to the home Component Directory.*
2. *Check the registration request state – checks the registration request state at the Component Directory and prints it to the console, updates Broker certificates when request state is approved. Note that the registration request state must be approved to continue with the next step.*
3. *Broker configuration push – pushes the Broker configuration to the Component Directory.*

To successfully finish the Broker registration process, Broker registration request must be approved by the administrator of the Component Directory. Please note that the outcome of the “Check registration request state” step can be approved, pending, or rejected. You can see the outcome in the console log output. The “Check registration request state” step can be executed multiple times.

The final step "*Broker configuration push*" can be executed successfully only if the outcome of the previous step "Check of registration request state" execution is approved (in other words the Broker registration request is approved by the Component Directory administrator and Broker certificates are updated).

To execute the commands, open the command prompt and change the current directory to */opt/ecp-broker/reg* where the broker-registration-tool.jar is located.

1. *Send the Broker registration requests by running the command:*

- ☐ *java -D"spring.config.location=/opt/ecp-broker/config/broker.properties" -jar broker-registration-tool.jar -m INITIAL_REGISTRATION*

2. *Check the status of the registration request by running the command:*

- ☐ *java -D"spring.config.location=/opt/ecp-broker/config/broker.properties" -jar broker-registration-tool.jar -m PROCESS_REGISTRATION*
- *To continue with next step the output from console should be:*
  - "*Request with ID {requestID} is approved. Storing signed AUTHENTICATION certificate*."
  - When console output is like: "Request with ID {requestID} is still pending..." or "Request with ID {requestID} was REJECTED by Component Directory administrator." please contact your Component Directory administrator for broker registration process assistance.

3. *Push the broker configuration to the Component Directory by running the command:*

- ☐ *java -D"spring.config.location=/opt/ecp-broker/config/broker.properties" -jar broker-registration-tool.jar -m PUSH_CONFIGURATION*

13.2.1. Broker registration tool proxy support

Broker registration tool can be used also for ECP networks accessible through forward proxy server. To enable proxy start tool with additional JVM command line parameters:

- *socksProxyHost – define proxy IP address or host name*
- *socksProxyPort – define proxy port*

For example, the Broker registration request can be sent through proxy server by running the following command:

- *java -D"spring.config.location=/opt/ecp-broker/config/broker.properties" -D"socksProxyHost=<HostIP>" -D"socksProxyPort=20080" -jar broker-registration-tool.jar -m INITIAL_REGISTRATION*

14. External Databases

To install ECP on an external database, follow a sub-chapter appropriate for your database system. The chapters list some ECP parameters related to the database that are specific for a given database system. For the complete list of parameters that must be configured, refer to the HA chapter for your component type:

- *Endpoint: Parameters required for HA installation*
- *Component Directory: Parameters required for HA installation*

Remember to configure your ECP instance for HA mode (profile) when using the external database even if you have the ECP installed on just one server.

14.1. MySQL database

14.1.1. Install the database

The supported versions of MySQL databases are 8.0 and 8.4. Please follow the official installation guides to install MySQL database for your environment.

Please be aware that there is a change from version 5.7 regarding root access to the server in a Linux environment. MySQL 5.7 or higher creates a password automatically and writes it into the error log which is by default located at: *n1 /var/log/mysql/error.log*

Example command to see temporary password:

- *tail -n1 /var/log/mysql/error.log*

Example output:

- *2016-09-20T11:32:17.262112Z 1 [Note] A temporary password is generated for root@localhost: t;/VkleMJ9eM*

14.1.2. Configure the database

ECP requires the following configuration in the mysql configuration file (usually located in /etc/**my.cnf**). It is recommended, but not mandatory, to change properties like *innodb-buffer-pool-size* and *innodb-log-file-size* according to the environment.

Properties required by ECP:

Parameter	Value

lower-case-table-names	1
max_allowed_packet	100M

14.1.3. Create application schema and configure user access

It is necessary to be able to connect from the ECP application servers under the application account to the MySQL database. The following command can be used to setup MySQL accounts. Please change the password included in this example:

- *mysql -u root -ppassword <<EOF*
- *CREATE USER IF NOT EXISTS 'ecp'@'%' IDENTIFIED BY 'ecp-password';*
- *CREATE DATABASE IF NOT EXISTS ecp*
- *DEFAULT CHARACTER SET utf8*
- *DEFAULT COLLATE utf8_general_ci;*
- *GRANT ALL ON ecp.* TO 'ecp'@'%';*
- *EOF*

Note: This command creates only an ECP user (used by application). When deploying the application in HA setup, user accounts and databases for Brokers should be created as well. A recommended database name is *ecp.*

14.1.4. ECP configuration changes

The following table shows MySQL-specific values for parameters that must be changed in the ecp/ecp-directory.properties configuration file.

Parameter	Value
spring.datasource.validation-query	select 1
ecp.db.url	jdbc:mariadb://<host>:<port>/<schema>?autoReconnect=true&characterEncoding=UTF-8 (If multi-line connection string is used, all lines, except the last one, must end with backslash)
spring.datasource.driver-class-name	org.mariadb.jdbc.Driver

14.2. MS SQL database

14.2.1. Install the database

The supported versions of MS SQL Database are SQL Server 2019 and SQL Server 2022. Other versions of the database may work correctly as well, but the support is not guaranteed and ECP is not tested against versions other than the supported one. Please follow the official installation guides to install SQL Server for your environment. The charset of the database instance is recommended to be set to NVARCHAR. Note that SQL Server Authentication mode must be set to mixed - SQL Server Authentication is required by ECP.

14.2.2. Create application databases

Microsoft SQL Server Management Studio is recommended for creation of ecp databases and users with logins.

The following T-SQL script snippets can be also used for ecp database creation.

A recommended database name is *ecp.*

14.2.2.1. Create database

Please check and modify database name and filenames before execution of the script.

- *USE [master]*
- *GO*
- *CREATE DATABASE [ecp]*
- *CONTAINMENT = NONE*
- *ON PRIMARY*
- *(NAME = N'ecp', FILENAME = N'<db_path>\ecp.mdf' , SIZE = 8192KB , MAXSIZE = UNLIMITED, FILEGROWTH = 65536KB)*
- *LOG ON*
- *(NAME = N'ecp_log', FILENAME = N'<db_path>\ecp_log.ldf' , SIZE = 73728KB , MAXSIZE = 2048GB , FILEGROWTH = 65536KB)*
- *GO*

14.2.2.2. Create login

Please check and modify the login name, password, and default database name before execution of the script.

- *USE [master]*
- *GO*
- *CREATE LOGIN [ecp] WITH PASSWORD=N'ecp-password', DEFAULT_DATABASE=[ecp], DEFAULT_LANGUAGE=[us_english], CHECK_EXPIRATION=OFF, CHECK_POLICY=OFF*
- *GO*

14.2.2.3. Create database user

Please check and modify the user and database name before execution of the script.

- *USE [ecp]*
- *GO*
- *CREATE USER [ecp] FOR LOGIN [ecp] WITH DEFAULT_SCHEMA=[dbo]*
- *GO*

14.2.2.4. Assign user to the following roles:

Please check and modify the user and database name before execution of the script.

- *USE [ecp]*
- *GO*
- *ALTER ROLE [db_datareader] ADD MEMBER [ecp]*
- *GO*
- *USE [ecp]*
- *GO*
- *ALTER ROLE [db_datawriter] ADD MEMBER [ecp]*
- *GO*
- *USE [ecp]*
- *GO*
- *ALTER ROLE [db_ddladmin] ADD MEMBER [ecp]*
- *GO*
- *USE [ecp]*
- *GO*
- *ALTER ROLE [db_owner] ADD MEMBER [ecp]*
- *GO*

14.2.2.5. Configure max degree of parallelism

It is recommended to disable parallelism by setting max degree of parallelism option to 1.

- *USE [ecp]*
- *GO*
- *EXEC sp_configure 'show advanced options', 1;*
- *GO*
- *RECONFIGURE WITH OVERRIDE;*
- *GO*
- *EXEC sp_configure 'max degree of parallelism', 1;*
- *GO*
- *RECONFIGURE WITH OVERRIDE;*
- *GO*

14.2.3. ECP configuration changes

The following table shows MSSQL-specific values for parameters that must be changed in the ecp/ecp-directory.properties configuration file.

Parameter	Value
spring.datasource.validation-query	select 1
ecp.db.url	jdbc:sqlserver://<host>:<port>;databaseName=<database> (If multi-line connection string is used, all lines, except the last one, must end with backslash)
spring.datasource.driver-class-name	com.microsoft.sqlserver.jdbc.SQLServerDriver

14.2.3.1. Windows authentication

If you wish to use Windows authentication instead of SQL Server authentication, add *integratedSecurity=true* option to the connection string in *ecp.db.url* property.

Download Microsoft JDBC Driver 12.8.1 for SQL Server from https://repo1.maven.org/maven2/com/microsoft/sqlserver/mssql-jdbc_auth/12.8.1.x64/mssql-jdbc_auth-12.8.1.x64.dll and copy the file to this directory:

- *<install_path>\tomcat\bin*

14.3. Oracle database

14.3.1. Install the database

The supported and tested version of Oracle Database is 19c. Other database versions may also work, but the support is not guaranteed and ECP is not tested against versions other than the supported one. Please follow the official installation guides to install Oracle Database for your environment.

The charset of the database instance is recommended to be set to AL32UTF8. If the database was already created with AL16UTF16 charset, following changes must be made in the ecp/ecp-directory.properties configuration file:

- *Add this line: spring.liquibase.parameters.oracle.useNationalCharacterSet=true*
- *Add defaultNChar and convertNcharLiterals parameters to the JDBC URL in ecp.db.url property*
    - *ecp.db.url=jdbc:oracle:thin:@<host>:<port>/<sid>?defaultNChar=true&convertNcharLiterals=true*

14.3.2. Create application databases

Oracle SQLDeveloper tool is recommended for creation of ECP databases and users with logins.

The following PL/SQL script snippets can be also used for ECP user and database creation.

A recommended database name is *ecp*.

14.3.2.1. Create application user and database

- *CREATE USER ecp IDENTIFIED BY password ;*

14.3.2.2. Assign user to the following roles and privileges

- *GRANT "CONNECT" TO ecp ;*
- *GRANT ALTER ANY INDEX TO ecp ;*
- *GRANT CREATE ANY INDEX TO ecp ;*
- *GRANT ALTER ANY TABLE TO ecp ;*
- *GRANT ALTER ANY SEQUENCE TO ecp ;*
- *GRANT CREATE TABLE TO ecp ;*

- *GRANT SELECT ANY SEQUENCE TO ecp ;*
- *GRANT MANAGE TABLESPACE TO ecp ;*
- *GRANT CREATE TABLESPACE TO ecp ;*
- *GRANT CREATE SEQUENCE TO ecp ;*
- *GRANT UPDATE ANY TABLE TO ecp ;*
- *GRANT ALTER TABLESPACE TO ecp ;*
- *GRANT READ ANY TABLE TO ecp ;*
- *GRANT INSERT ANY TABLE TO ecp ;*
- *GRANT ALTER ANY TYPE TO ecp ;*
- *GRANT CREATE ANY TABLE TO ecp ;*
- *GRANT ALTER DATABASE TO ecp ;*
- *GRANT DROP TABLESPACE TO ecp ;*

The following command sets unlimited tablespace. If this is not desired behaviour, please contact your DBA to set up quotas for you.

- *GRANT UNLIMITED TABLESPACE TO ecp ;*

14.3.3. ECP configuration changes

The following table shows Oracle-specific values for parameters that must be changed in the ecp/ecp-directory.properties configuration file.

Parameter	Value
spring.datasource.validation-query	select 1 from dual
ecp.db.url	jdbc:oracle:thin:@<host>:<port>/<sid> (If multi-line connection string is used, all lines, except the last one, must end with backslash)
spring.datasource.driver-class-name	oracle.jdbc.OracleDriver

14.4. PostgreSQL database

14.4.1. Install the database

The supported versions of PostgreSQL Database are 13 and 16. Other database versions may also work, but the support is not guaranteed and ECP is not tested against versions other than the supported ones. Please follow the official installation guides to install PostgreSQL database for your environment.

14.4.2. Create application databases

PostgreSQL *pgAdmin* is the recommended tool for creation of ECP databases and users with logins.

The following SQL script snippets can also be used for ECP user and the database creation.

A recommended database name is *ecp*.

14.4.2.1. Create application user and database

- *CREATE ROLE ecp LOGIN PASSWORD 'ecp-password'*
- *NOSUPERUSER INHERIT NOCREATEDB NOCREATEROLE NOREPLICATION;*
- *CREATE DATABASE ecp*
- *WITH OWNER = ecp*
- *ENCODING = 'UTF8'*
- *TABLESPACE = pg_default*
- *CONNECTION LIMIT = -1;*

14.4.3. ECP configuration changes

The following table shows PostgreSQL-specific values for parameters that must be changed in the ecp/ecp-directory.properties configuration file.

Parameter	Value
spring.datasource.validation-query	select 1
ecp.db.url	jdbc:postgresql://<host>:<port>/<database> (If multi-line connection string is used, all lines, except the last one, must end with backslash)
spring.datasource.driver-class-name	org.postgresql.Driver

15. Docker Support

ECP components can be run within Docker containers – ECP components are shipped also as Docker images. We also provide Docker Compose file for a setup of the basic test ECP network.

Using ECP in Docker assumes basic Docker skills. To get familiar with Docker please visit following links:

- *https://docs.docker.com/engine/docker-overview/*
- *https://docs.docker.com/*

15.1. Prerequisites

- *Docker for Windows / Linux installed. Please see https://docs.docker.com/engine/installation/ for instructions.*
- *Access to ECP repositories on Docker Hub (https://hub.docker.com/) – can be granted on request to your existing Docker Hub account*
    - *Alternatively, ECP Endpoint and Internal Broker can be obtained from the ENTSO-E ECP Download page instead (https://www.entsoe.eu/ecco-sp/ecp/)*

15.2. ECP Docker images

ECP is delivered with Docker Images. The ECP Docker Images are available for all ECP components (Component Directory, Endpoint, Internal Broker, and Broker). Images are available at Docker Hub (https://hub.docker.com/) under the following names:

- *entsoe/ecp-directory*
- *entsoe/ecp-endpoint*
- *entsoe/eccosp-artemis*
- *entsoe/ecp-broker*

Images are tagged with their version, and you can always get the latest revision of selected minor version by pulling tag containing only major and minor versions.

ECP Endpoint and Internal Broker Docker images (provided as tar files) can also be downloaded from the ENTSO-E ECP Download page, by selecting *"Docker Binary"* in the *"Choose Package Type"* section: https://www.entsoe.eu/ecco-sp/ecp/

The *docker load* command is used to load an image from the tar file.

ECP Docker images are digitally signed. To enable digital signature verification on the client side, Docker Content Trust must be enabled by setting the value of *DOCKER_CONTENT_TRUST* environmental variable to *1*. With Docker Content Trust enabled, Docker images cannot be pulled from Docker Hub or started when they are not successfully verified.

Before you start your container with ECP software, check the following sections and consider the use of port and volume mapping. It is recommended to store your data outside the Docker container either by mounting a data directory on your Docker host to the volume inside the container (see the section) or by using an external database.

To avoid problems with a connection between ECP containers, please do not use underscores in ECP container name.

15.2.1. Exposed ports

ENDPOINT

- *8443 – provides access to web GUI*

INTERNAL BROKER

- *8161 – provides access to the Artemis web management console*
- *5671 – optional AMQPS connector used for direct communication between Endpoints*
- *5672 – AMQPS connector used for the Endpoint and Business Applications connections*

COMPONENT DIRECTORY

- *8443 – provides access to web GUI and to the Component Directory API (e.g., for the synchronization)*

BROKER

- *8161 – provides access to the Artemis web management console*
- *5671 – AMQPS connector for the Endpoint connections*

15.2.2. Environment variables

Endpoint and Component Directory:

- *CATALINA_OPTS – this variable can be used to modify JVM memory allocation for Endpoint or Directory. Also, additional JVM parameters can be specified.*

15.3. ECP Docker Compose test environment

Together with ECP Docker Images, the Docker Compose test environment is also provided.

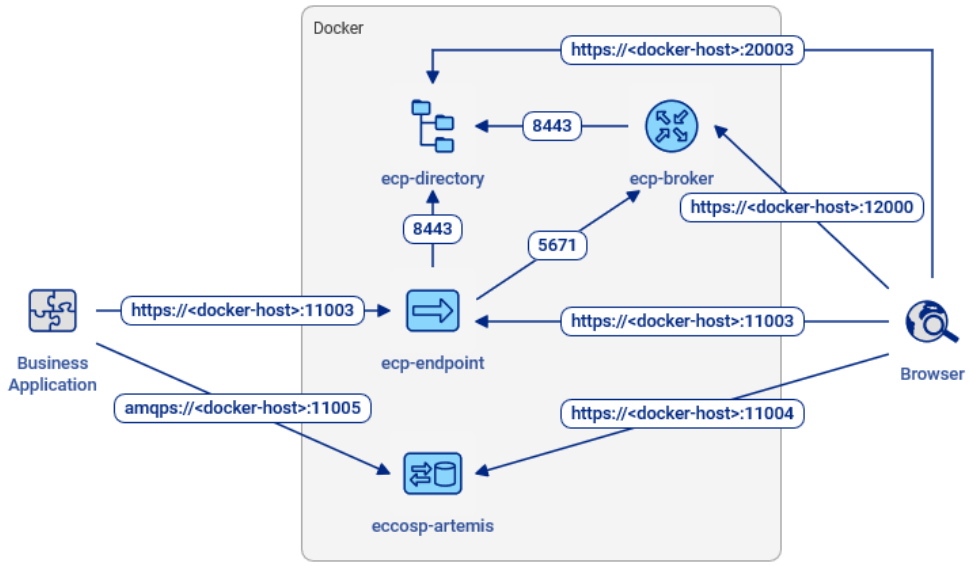

Figure 41: Docker Compose test environment

The test environment can be easily run with one CLI command and contains all available ECP component types (Component Directory, Endpoint, Internal Broker, Broker), one container instance per component type.

15.3.1. How to start / stop Docker Compose environment

Open command line and change to folder where test environment file ***docker-compose.yml*** is located. Do not forget to set folder permissions. Next execute command:

- *docker compose up -d*

Note: Flag *-d* is optional, it will run containers in the background.

Wait until all ECP containers are running. For testing purposes, all ECP components are already registered under the started Component Directory, therefore manual component registration is not required. The Endpoint has also a default message path configured so you can immediately start sending messages.

To check if all components are running execute Docker CLI command:

- *docker ps*

You can also open browser and visit URLs:

- *https://127.0.0.1:20003/ECP_MODULE for access to Component Directory (login admin@password)*
- *https://127.0.0.1:11003/ECP_MODULE for access to Endpoint*
- *https://127.0.0.1:11004 for access to Internal Broker Artemis management console (login endpoint@password)*
- *https://127.0.0.1:12000 for access to ECP Broker Artemis management console (login admin@password)*

If you are not running the Docker on your machine, substitute the IP address 127.0.0.1 for the address of the Docker host machine.

Once the environment is started, you can use it for your testing purposes, e.g., to evaluate the platform or to test the integration with your business application. By default, the Endpoint has only Web Service interface enabled, but you can configure the Endpoint however you need using the standard procedure (see the Administration Guide).

To stop ECP Docker test environment, either press “Ctrl+C” keyboard shortcut within the terminal running the “docker compose up -d” command in foreground, or issue command “*docker compose stop*” under the folder containing the ***docker-compose.yml*** file.

15.3.2. ECP Docker Compose environment configuration

15.3.2.1. Logs

Endpoint and Component Directory components log files are in the following folders:

- *directory/logs*
- *endpoint/logs*
- *artemis/logs*
- *broker/logs*

15.3.2.2. ECP component configuration

Component configuration can be changed manually by editing its property file. The property file is located at nested config folder under the component root folder:

- *directory/config/**
- *endpoint/config/**
- *artemis/config/**
- *broker/config/**

For instructions, how to configure specific component please see the Administration Guide.

15.3.2.3. ECP containers configuration

Delivered testing Docker compose configuration (***docker-compose.yml***) can be edited manually to experiment with the ECP setup. You can change the default port mapping for containers, specify additional JVM parameters for ECP components (e.g., limit the memory settings) or you can also add an additional Endpoint to the network.

Please see the Docker Compose documentation online (https://docs.docker.com/compose/) for more information on the configuration file options.

15.3.2.4.Env file

Together with **docker-compose.yml** file the **.env** file is also provided. In **.env** file you can define version (tag) of ECP/EDX which will be used. There are two possible values. Value *latest*, which will use the latest available version of ECP/EDX or specific version (tag). Version can be also specified in **docker-compose.yml** itself. It can be defined in *image* property (*image: "entsoe/ecp-endpoint:<ECP version>"*).

Example:

- *ECP_VERSION=latest*

or

- *ECP_VERSION=<ECP version>*

15.3.2.5.How to add another component

New component needs to be added in **docker-compose.yml**. Please note, that in case of ECP Endpoint, adding the Internal Broker (eccosp-artemis) is required. In compose, copy some existing component, and change following values (values must be unique):

- *Service name*
- *container_name*
- *volumes*
- *ports*
- *ipv4_address*

In section *volumes*, delete (comment out) link with data and config folders. They will be linked later. Then create folder for new component, make sure that the new folders have 777 permissions.

Run *docker compose up -d* command to start all containers and check if your newly added component is running (use *docker ps* to list all active containers).

Use *docker cp <container>:<src_path> <dest_path>* to copy files (all configuration files, database, etc.) from container to local file system and link them with container in **docker-compose.yml**, section *volumes* (make sure that the copied files have 777 permissions). After that, run following commands:

- *docker compose stop [service code]*
- *docker compose rm [service code]*
- *docker compose up -d [service code]*

Newly added component should now use configuration from local file system.

15.3.3. Useful Docker commands

Command	Description
docker login	Log in to docker.
docker ps	List all active containers.
docker system prune	Remove all stopped containers, all volumes not used by at least one container, all networks not used by at least one container, all dangling images.
docker exec -it <container ID> /bin/bash	Access container terminal.
docker cp <container>:<src_path> <dest_path> docker cp <src_path> <container>:<dest_path>	Copy file from/to container.
docker compose pull	Pull an image or a repository from a registry.
docker compose up -d [service name]	Start container in the background. When [service name] is missing, all containers defined in **docker-compose.yml** will be started.
docker stop [container ID or name]	Stop container.
docker restart [container ID or name]	Restart container.

docker compose stop [service name]	Stop container. When [service name] is missing, all active containers defined in **docker-compose.yml** will be stopped.
docker compose restart [service name]	Restart container. When [service code] is missing, all active containers defined in **docker-compose.yml** will be restarted.
docker compose rm [service name]	Remove container. Container must be stopped.

16. Kubernetes Support

ECP installer packages provide Helm charts and Values files. To learn how to deploy ECP components in the Kubernetes cluster, please see the ECP Kubernetes Guide.

17. Component Deployment - Security Recommendations

This section provides information about secure deployment of ECP components. The following recommendations represent the recommended level of security, which can be further enhanced (e.g. by additional components like proxies if necessary).

17.1. Endpoint deployment

ECP Endpoint secure deployment is formed of high availability (HA) deployment and managed access rules. Access to Endpoint cluster is not restricted inside demilitarized zone (DMZ). Outside DMZ inbound access to Endpoint is completely denied by the firewall rules. Outbound communication from Endpoint to broker or Component Directory is allowed at firewall for AMQPS and HTTPS protocol ports e.g.: 5671, 8443. HTTP access to the Endpoint cluster nodes can be routed through a reverse proxy inside DMZ to provide load balancing and transparent failover. This applies for business applications using web service API and for the users accessing ECP via GUI.

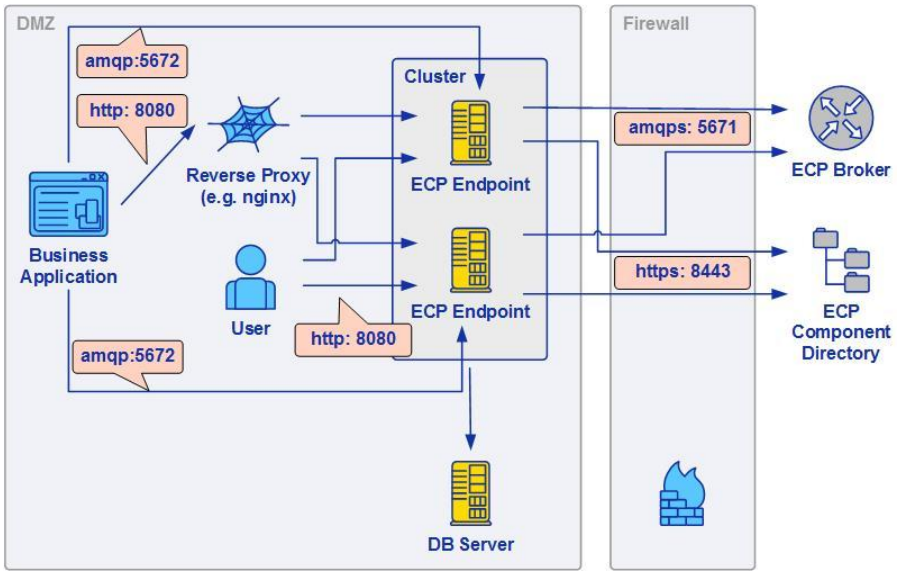

Figure 42: Endpoint Deployment

17.2. Component Directory deployment

Component Directory not only opens outbound connections, but it must also be able to accept inbound connections from other ECP components. Inbound connections are allowed only for HTTPS protocol (default ECP port setting is 8443). HTTPS protocol is used by Component Directory synchronization. Just like Endpoint, Component Directory is recommended to be deployed at HA setup – access to cluster nodes is load balanced (reverse proxy).

HTTP Reverse Proxy cannot be used due to Authentication Certificate that needs to be validated on the Component Directory. HTTP Reverse Proxy normally terminates the SSL connection and removes the client Authentication Certificate.

The deployment was tested against NGINX proxy.

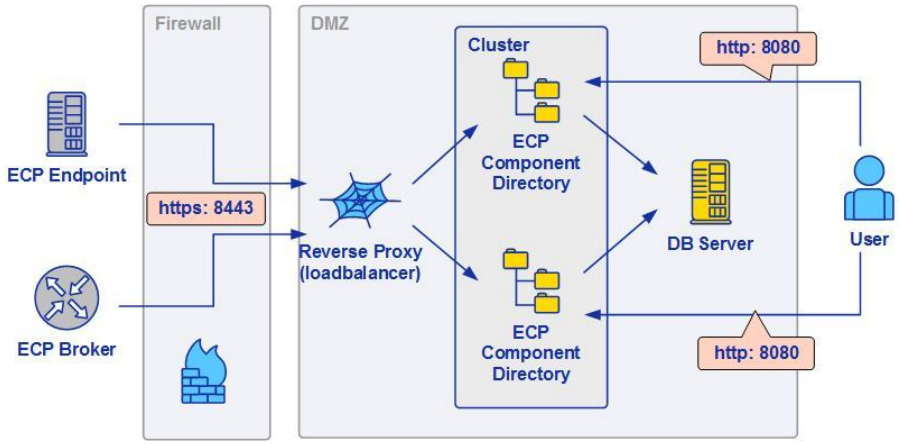

Figure 43: Component Directory deployment

17.3. Broker deployment

Outbound connections from the Broker are allowed only for Component Directory synchronization on HTTPS protocol (port 8443). Inbound connections from Endpoints are allowed for AMQPS protocol (default ECP port setting is 5671). Just like Endpoint and Component Directory, Broker is recommended to be deployed at HA setup. Access to the Broker cluster nodes can be provided through TCP load balancer / reverse proxy or it can be addressed directly from Endpoint to the specific Broker node through configured multiple failover URLs.

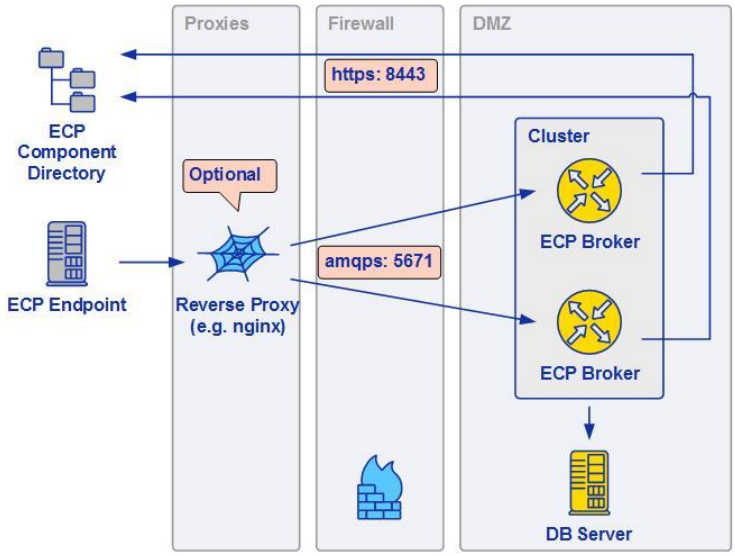

Figure 44: Broker deployment

18. FAQ

This section contains the most frequently asked questions and answers to these questions in a single place.

18.1. Support of a forward proxy

Question: Is it possible to use http proxy for ECP?

Answer: ECP uses the AMQP protocol to exchange data and it is not possible to use http proxy. It is possible to use SOCKS5 proxy, which works on a lower session layer of OSI network model and is agnostic to the application protocol.

ECP was tested with Dante proxy server (https://www.inet.no/dante/). See the Administration Guide for details on the configuration of proxy in ECP.

18.2. Support of a reverse proxy

Question: Is it possible to use a reverse proxy for inbound connections?

Answer: Yes, it is possible with some constraints put by the transport protocols, specifically AMQP. The reverse proxy directing traffic to a Broker or Endpoint must support a TCP forwarding to handle AMQP connections. The Component Directory uses only HTTP communication protocol, so it is possible to use any HTTP proxy or the TCP forwarding as well.

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ECP was tested with NGINX (https://www.nginx.com) to intercept the inbound connections using its TCP and UDP load balancing features (https://www.nginx.com/resources/admin-guide/tcp-load-balancing/). Below is a basic sample configuration to proxy connections to an ECP Broker.

```
stream {
    server {
        listen    5671;
        proxy_pass stream_brokers;
    }
    upstream stream_brokers {
        server 172.22.0.11:5671;
    }
}
```

18.3. Support of other databases for ECP Endpoint

Question: Is it possible to use other than the supported databases?

Answer: For non-HA mode, it is possible to use an embedded database, which does not have to be installed. For HA mode, it is only possible to use one of the supported external databases listed in chapter *External Databases*.

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